

Discrete-Time Signals: Time-Domain Representation (1/10)

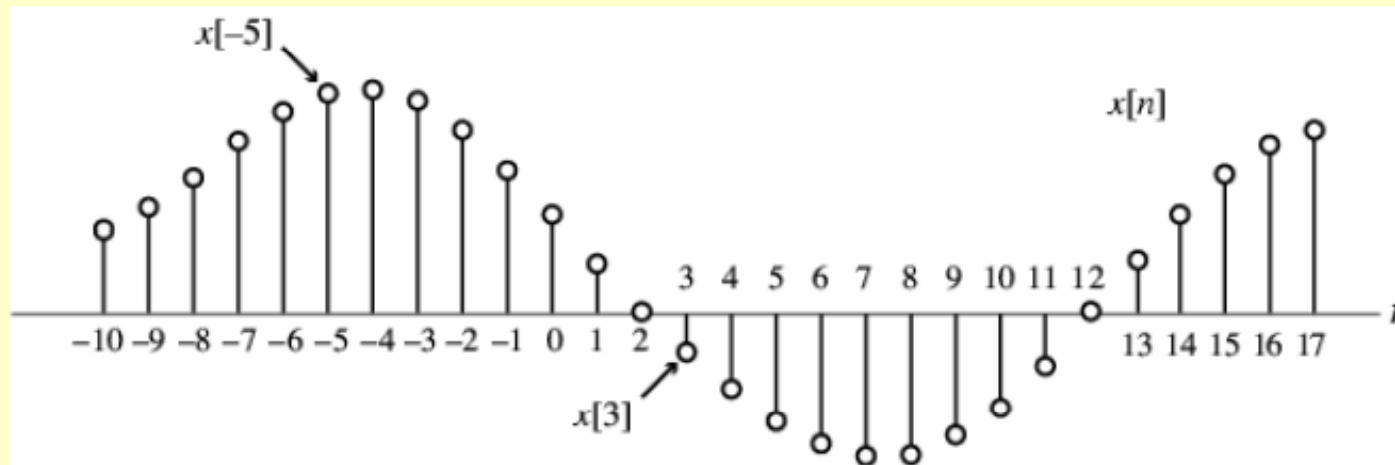
- Signals represented as sequences of numbers, called **samples**
- Sample value of a typical signal or sequence denoted as $x[n]$ with n being an integer in the range $-\infty \leq n \leq \infty$
- $x[n]$ defined only for integer values of n and undefined for non-integer values of n
- Discrete-time signal represented by $\{x[n]\}$
- Discrete-time signal may also be written as a sequence of numbers inside braces:

$$\{x[n]\} = \{\dots, -0.2, 2.2, 1.1, 0.2, -3.7, 2.9, \dots\}$$

↑

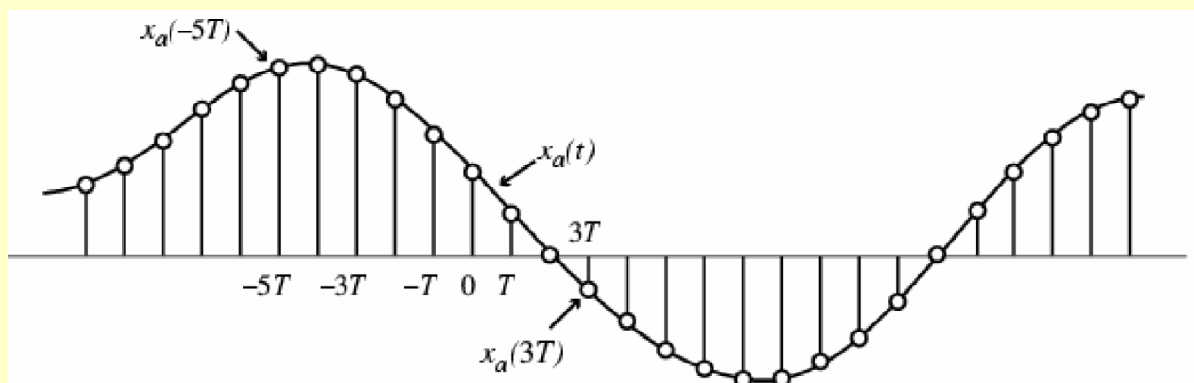
Discrete-Time Signals: Time-Domain Representation (2/10)

- Graphical representation of a discrete-time signal with real-valued samples is as shown below:



Discrete-Time Signals: Time-Domain Representation (3/10)

- In some applications, a discrete-time sequence $\{x[n]\}$ may be generated by periodically sampling a continuous-time signal $x_a(t)$ at uniform intervals of time



- Here, the n -th sample is given by

$$x[n] = x_a(t)|_{t=nT} = x_a(nT), \quad n = \dots, -2, -1, 0, 1, \dots$$

Discrete-Time Signals: Time-Domain Representation (4/10)

- The spacing T between two consecutive samples is called the **sampling interval** or **sampling period**
- Reciprocal of sampling interval T , denoted as F_T , is called the **sampling frequency** (in Hz)

$$F_T = \frac{1}{T}$$

- A complex sequence $\{x[n]\}$ can be written as $\{x[n]\} = \{x_{re}[n]\} + j\{x_{im}[n]\}$ where $x_{re}[n]$ and $x_{im}[n]$ are the real and imaginary parts of $x[n]$
- Often the braces are ignored to denote a sequence if there is no ambiguity

Discrete-Time Signals: Time-Domain Representation (5/10)

- Example - $\{x[n]\} = \{\cos 0.25n\}$ is a real sequence
- $\{y[n]\} = \{e^{j0.3n}\}$ is a complex sequence
- We can rewrite

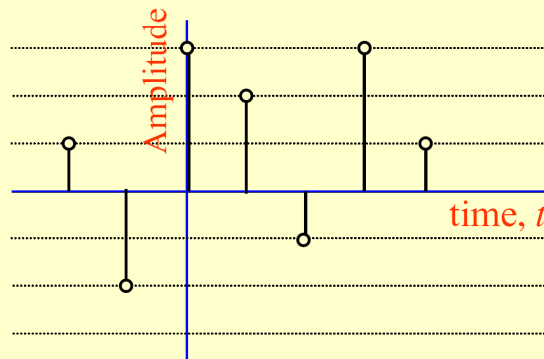
$$\begin{aligned}\{y[n]\} &= \{\cos 0.3n + j \sin 0.3n\} \\ &= \{\cos 0.3n\} + j\{\sin 0.3n\}\end{aligned}$$

where

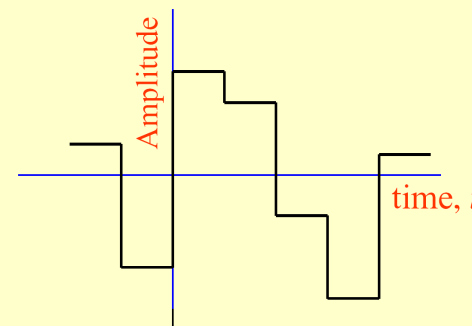
$$\begin{aligned}\{y_{re}[n]\} &= \{\cos 0.3n\} \\ \{y_{im}[n]\} &= \{\sin 0.3n\}\end{aligned}$$

Discrete-Time Signals: Time-Domain Representation (6/10)

- Two types of discrete-time signals:
 - **Sampled-data signals** in which samples are continuous-valued
 - **Digital signals** in which samples are discrete-valued (by quantizing the sample values either by **rounding** or **truncation**)



Digital signal



Boxedcar signal

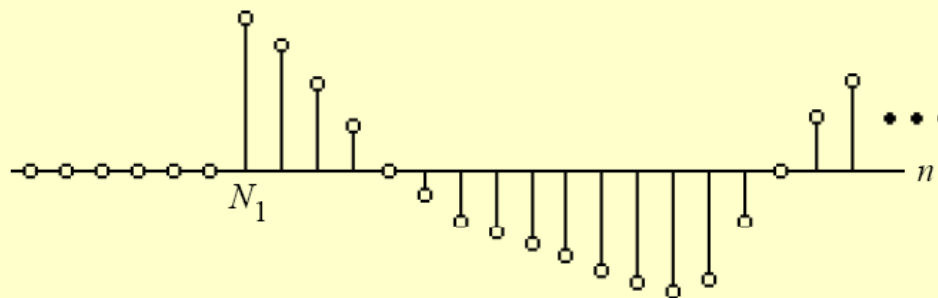
Discrete-Time Signals: Time-Domain Representation (7/10)

- A discrete-time signal may be a **finite-length** or an **infinite-length sequence**
- Example - $x[n] = n^2$, $-3 \leq n \leq 4$ is a finite-length sequence of length 8
- $y[n] = \cos 0.4n$ is an infinite-length sequence
- A length- N sequence is often referred to as an **N -point sequence**
- The length of a finite-length sequence can be increased by **zero-padding**, i.e., by appending it with zeros

$$x_e[n] = \begin{cases} n^2, & -3 \leq n \leq 4 \\ 0, & 5 \leq n \leq 8 \end{cases}$$

Discrete-Time Signals: Time-Domain Representation (8/10)

- A **right-sided sequence** $x[n]$ has zero-valued samples for $n < N_1$



A right-sided sequence

- If $N_1 \geq 0$, a right-sided sequence is called a **causal sequence**
- A **left-sided sequence** $x[n]$ has zero-valued samples for $n > N_2$ (called a **anti-causal sequence** if $N_2 \leq 0$)

Discrete-Time Signals: Time-Domain Representation (9/10)

- **Size of a Signal** - given by the norm of the signal

L_p -norm :

$$\|x\|_p = \left(\sum_{n=-\infty}^{\infty} |x[n]|^p \right)^{1/p}$$

where p is a positive integer

- The value of p is typically 1 or 2 or ∞

L_2 -norm $\|x\|_2$ is the **root-mean-squared (rms) value** of $\{x[n]\}$

L_1 -norm $\|x\|_1$ is the **mean absolute value** of $\{x[n]\}$

L_∞ -norm $\|x\|_\infty$ is the **peak absolute value** of $\{x[n]\}$ (why?)

$$\|x\|_\infty = |x|_{\max}$$

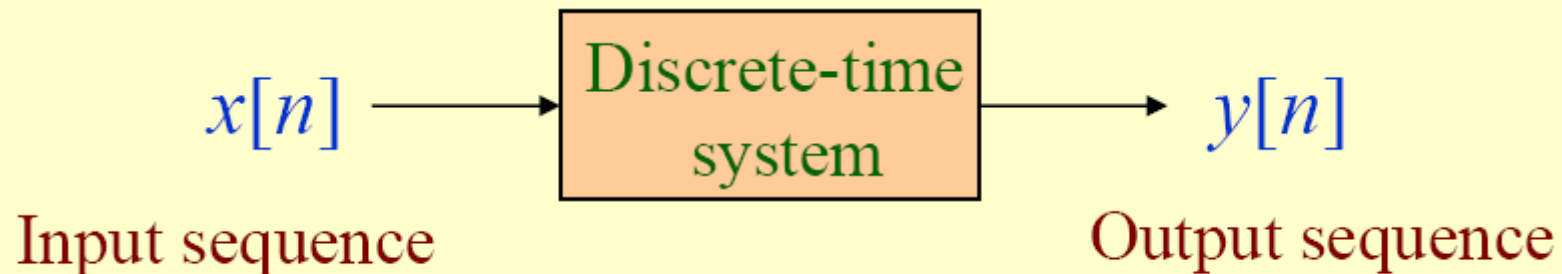
Discrete-Time Signals: Time-Domain Representation (10/10)

Example -

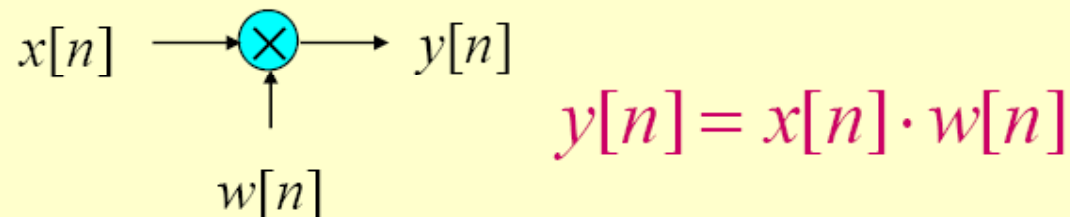
- Let $\{y[n]\}$, $0 \leq n \leq N-1$, be an approximation of $\{x[n]\}$, $0 \leq n \leq N-1$
- An estimate of the relative error is given by the ratio of the L_2 -norm of the difference signal and the L_2 -norm of $\{x[n]\}$:

$$E_{rel} = \left(\frac{\sum_{n=0}^{N-1} |y[n] - x[n]|^2}{\sum_{n=0}^{N-1} |x[n]|^2} \right)^{1/2}$$

Elementary Operations on Sequences



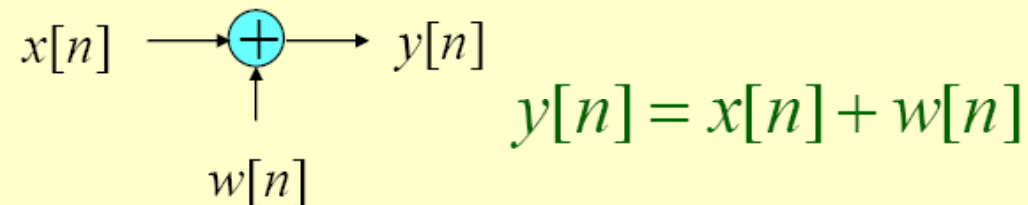
- **Product (modulation)** operation:



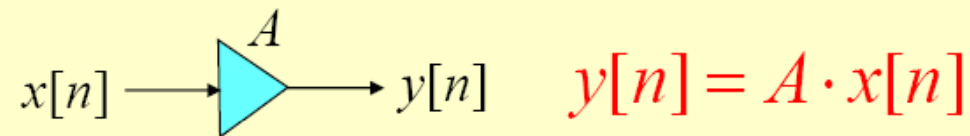
- An application is in forming a finite-length sequence from an infinite-length sequence by multiplying the latter with a finite-length sequence called an **window sequence (windowing)**

Elementary Operations on Sequences

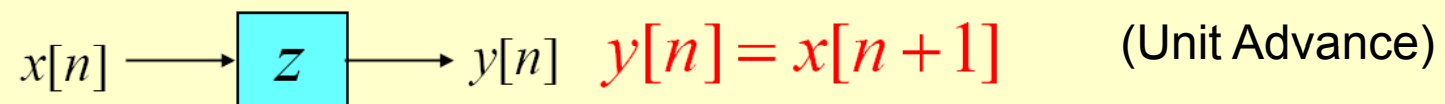
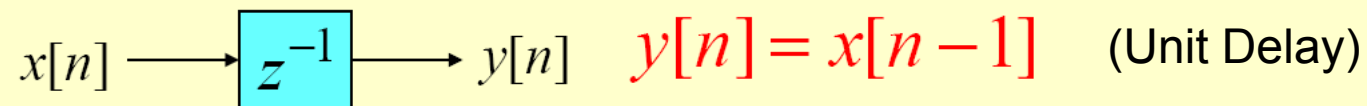
- **Addition** operation:



- **Multiplication** operation:



- **Time-shifting** operation: $y[n] = x[n - M]$

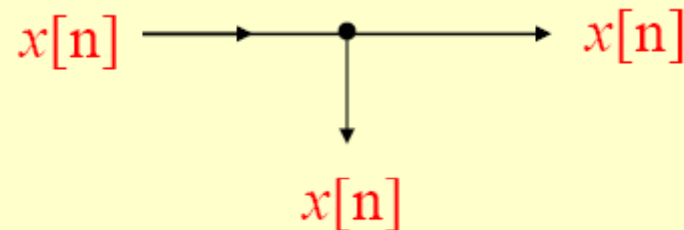


Elementary Operations on Sequences

- **Time-reversal (folding)** operation:

$$y[n] = x[-n]$$

- **Branching** operation: used to provide multiple copies of a sequence



- Example:

$$\{c[n]\} = \{a[n] \cdot b[n]\} = \{6 \quad -4 \quad 24 \quad -45 \quad 0\}$$

$$\{d[n]\} = \{a[n] + b[n]\} = \{5 \quad 3 \quad 10 \quad -4 \quad -3\}$$

$$\{e[n]\} = \frac{3}{2}\{a[n]\} = \{4.5 \quad 6 \quad 9 \quad -13.5 \quad 0\}$$

Elementary Operations on Sequences

Ensemble Averaging

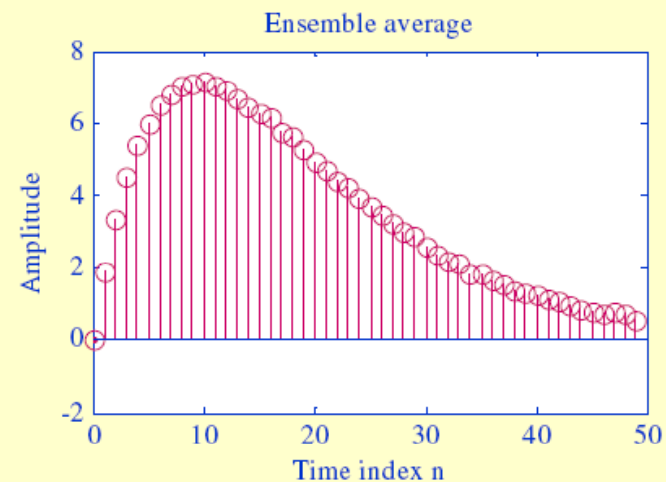
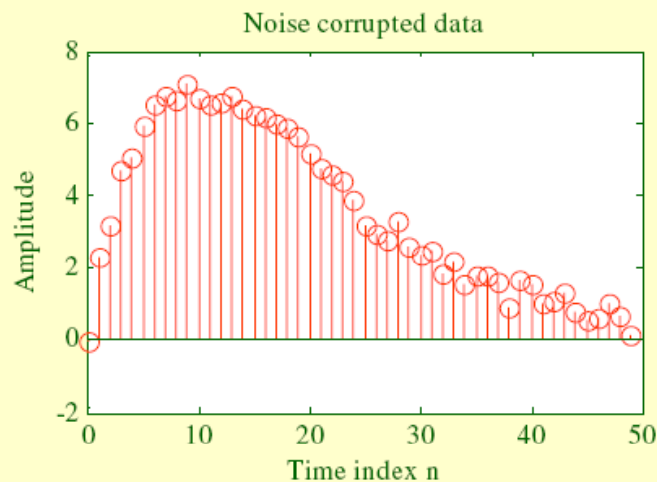
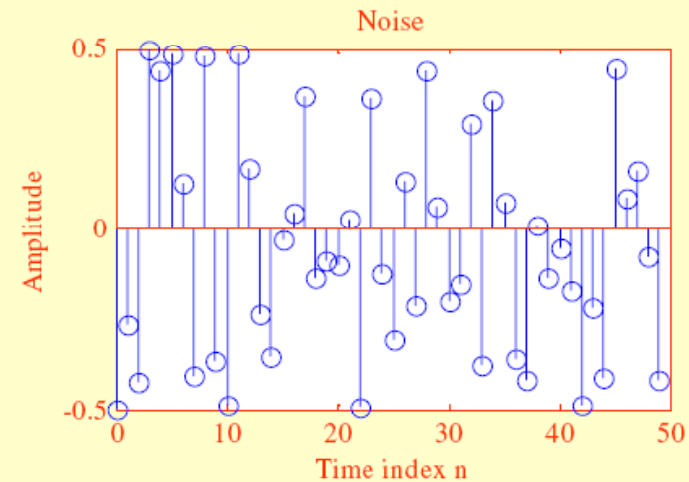
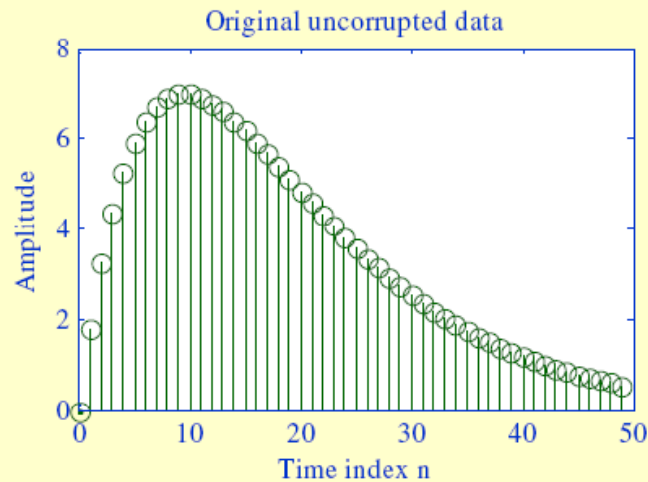
- An application of the addition operation in improving the quality of measured data corrupted by an additive random noise
- Let $\mathbf{d}_i$ denote the noise vector corrupting the i -th measurement of the uncorrupted data vector $\mathbf{s}$

$$\mathbf{x}_i = \mathbf{s} + \mathbf{d}_i$$

- The average data vector, called the ensemble average, obtained after K measurements is given by

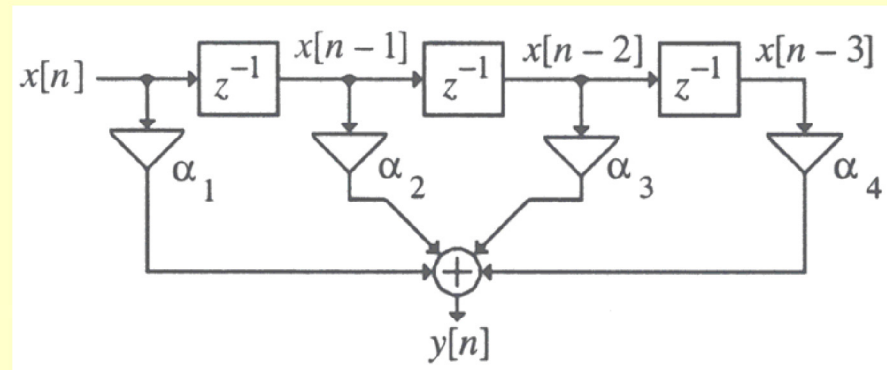
$$\mathbf{x}_{ave} = \frac{1}{K} \sum_{i=1}^K \mathbf{x}_i = \frac{1}{K} \sum_{i=1}^K (\mathbf{s} + \mathbf{d}_i) = \mathbf{s} + \frac{1}{K} \sum_{i=1}^K \mathbf{d}_i$$

Elementary Operations on Sequences



Combinations of Basic Operations

- Example -



$$y[n] = \alpha_1 x[n] + \alpha_2 x[n-1] + \alpha_3 x[n-2] + \alpha_4 x[n-3]$$

Sampling Rate Alteration

- A process to generate a new sequence $y[n]$ with a sampling F'_T rate higher or lower than that of the sampling rate F_T of a given sequence $x[n]$
- **Sampling rate alteration ratio:**

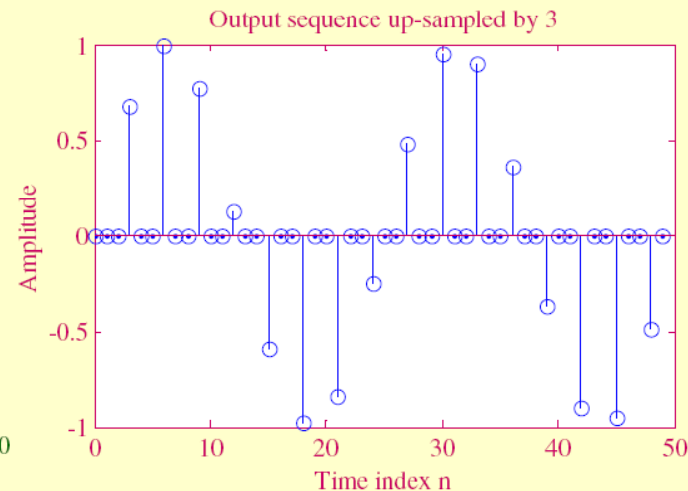
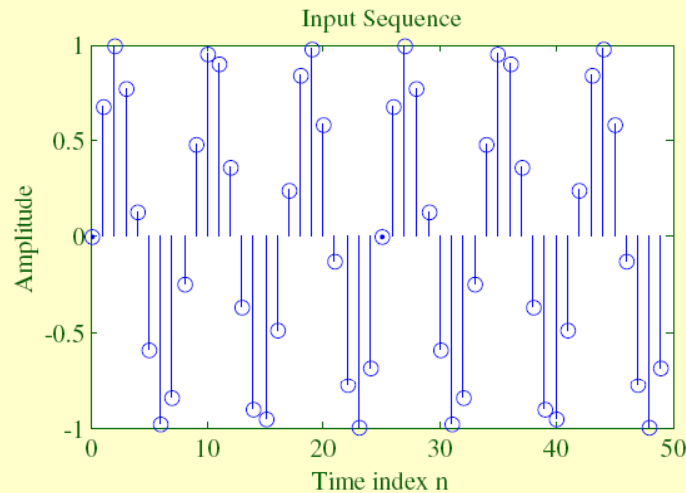
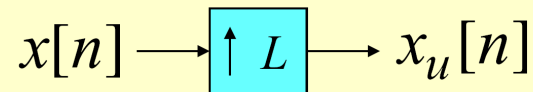
$$R = \frac{F'_T}{F_T}$$

- if $R > 1$, the process called interpolation
- if $R < 1$, the process called decimation

Up-Sampling

- In **up-sampling** by an integer factor $L > 1$, $L - 1$ equidistant zero-valued samples are inserted between each two consecutive samples of the input sequence $x[n]$:

$$x_u[n] = \begin{cases} x[n/L], & n = 0, \pm L, \pm 2L, \dots \\ 0, & \text{otherwise} \end{cases}$$

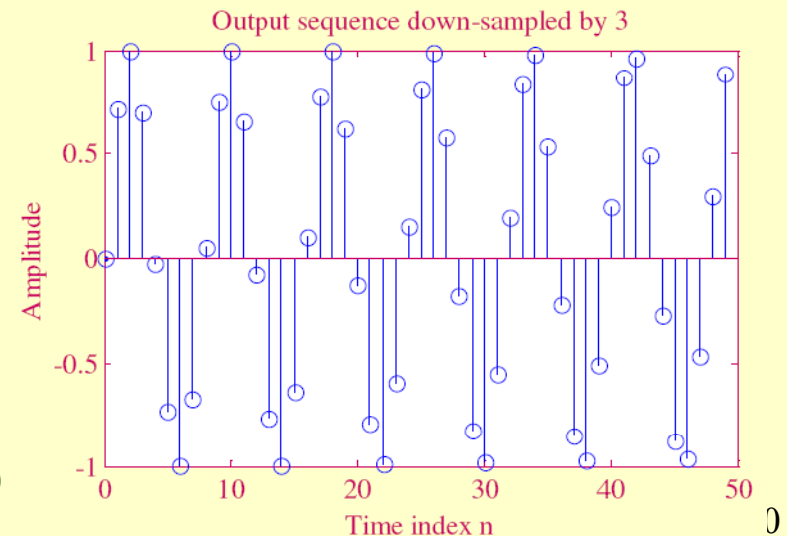
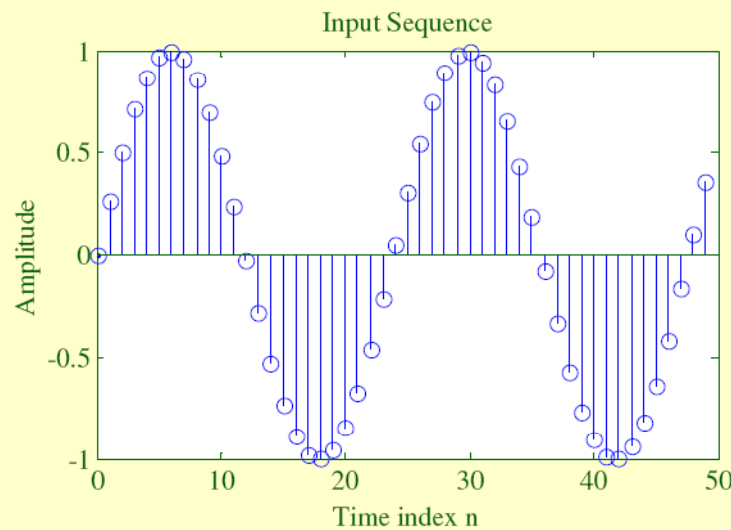


Down-Sampling

- In **down-sampling** by an integer factor $M > 1$, every M -th samples of the input sequence are kept and $M - 1$ in-between samples are removed:

$$y[n] = x[nM]$$

$$x[n] \rightarrow \boxed{\downarrow M} \rightarrow y[n]$$

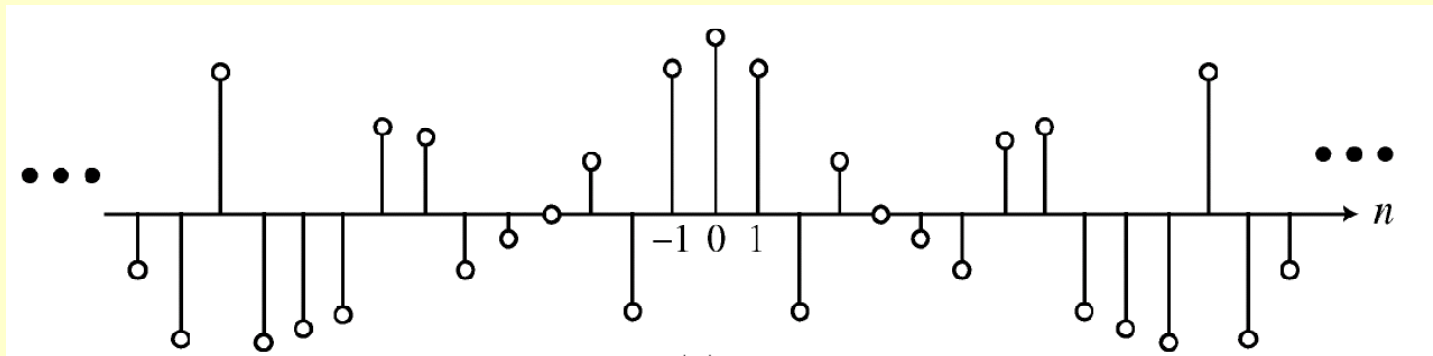


Classification of Sequences Based on Symmetry (1/4)

- **Conjugate-symmetric** sequence:

$$x[n] = x^*[-n]$$

- If $x[n]$ is real, then it is an **even sequence**
- for a conjugate-symmetric sequence $\{x[n]\}$, $x[0]$ must be a real number

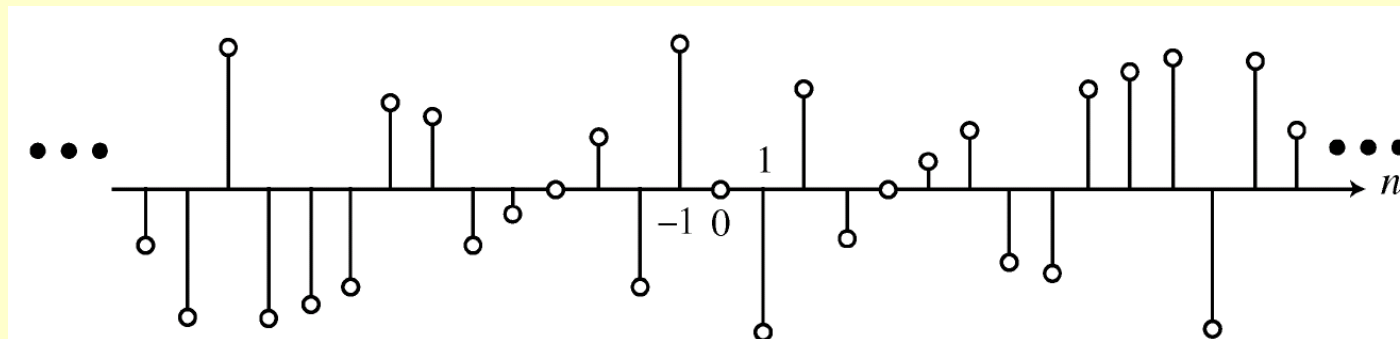


Classification of Sequences Based on Symmetry (2/4)

- **Conjugate-antisymmetric** sequence:

$$x[n] = -x^*[-n]$$

- If $x[n]$ is real, then it is an **odd sequence**
- for a conjugate anti-symmetric sequence $\{y[n]\}$, $y[0]$ must be an imaginary number



Classification of Sequences Based on Symmetry (3/4)

- Any complex sequence can be expressed as a sum of its conjugate-symmetric part and its conjugate-antisymmetric part:

$$x[n] = x_{cs}[n] + x_{ca}[n]$$

where

$$x_{cs}[n] = \frac{1}{2}(x[n] + x^*[-n])$$

$$x_{ca}[n] = \frac{1}{2}(x[n] - x^*[-n])$$

- Consider the length-7 sequence defined for $-3 \leq n \leq 3$

$$\{g[n]\} = \{0, 1+j4, -2+j3, 4-j2, -5-j6, -j2, 3\}$$

$$\{g_{cs}[n]\} = \{1.5, 0.5+j3, -3.5+j4.5, 4, -3.5-j4.5, 0.5-j3, 1.5\}$$

$$\{g_{ca}[n]\} = \{-1.5, 0.5+j, 1.5-j1.5, -j2, -1.5-j1.5, -0.5-j, 1.5\}$$

Classification of Sequences Based on Symmetry (4/4)

- Any real sequence can be expressed as a sum of its even part and its odd part:

$$x[n] = x_{ev}[n] + x_{od}[n]$$

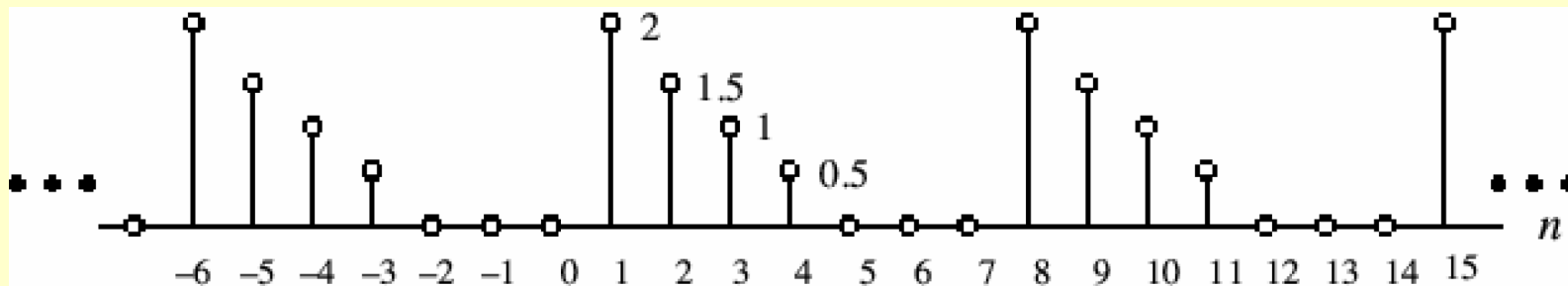
where

$$x_{ev}[n] = \frac{1}{2}(x[n] + x[-n])$$

$$x_{od}[n] = \frac{1}{2}(x[n] - x[-n])$$

Classification of Sequences Based on Periodicity

- A sequence $\tilde{x}[n]$ satisfying $\tilde{x}[n] = \tilde{x}[n + kN]$ is called a periodic sequence with a period N where N is a positive integer and k is any integer
- Smallest value of N satisfying $\tilde{x}[n] = \tilde{x}[n + kN]$ is called the **fundamental period**



- A sequence not satisfying the periodicity condition is called an **aperiodic sequence**

Classification of Sequences

Energy & Power Signals (1/3)

- Total **energy** of a sequence $x[n]$ is defined by

$$\mathcal{E}_x = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

- An infinite length sequence with finite sample values may or may not have finite energy
- The **average power of an aperiodic** sequence is defined by

$$P_x = \lim_{K \rightarrow \infty} \frac{1}{2K+1} \sum_{n=-K}^K |x[n]|^2$$

- Define the **energy** of a sequence $x[n]$ over a finite interval $-K \leq n \leq K$ as

$$\mathcal{E}_{x,K} = \sum_{n=-K}^K |x[n]|^2 \qquad P_x = \lim_{K \rightarrow \infty} \frac{1}{2K+1} \mathcal{E}_{x,K}$$

Classification of Sequences

Energy & Power Signals (2/3)

- The **average power** of a periodic sequence with a period N is given by

$$P_x = \frac{1}{N} \sum_{n=0}^{N-1} |\tilde{x}[n]|^2$$

- The average power of an infinite-length sequence may be finite or infinite
- Example - Consider the causal sequence defined by

$$x[n] = \begin{cases} 3(-1)^n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

- Note: $x[n]$ has infinite energy, its average power is

$$P_x = \lim_{K \rightarrow \infty} \frac{1}{2K+1} \left(9 \sum_{n=0}^K 1 \right) = \lim_{K \rightarrow \infty} \frac{9(K+1)}{2K+1} = 4.5$$

Classification of Sequences

Energy & Power Signals (3/3)

- An infinite energy signal with finite average power is called a **power signal**
 - Example - A periodic sequence which has a finite average power but infinite energy
- A finite energy signal with zero average power is called an **energy signal**
 - Example - A finite-length sequence which has finite energy but zero average power

Other Types of Classification (1/2)

- A sequence $x[n]$ is said to be **bounded** if ◦

$$|x[n]| \leq B_x < \infty$$

- Example - The sequence $x[n] = \cos 0.3\pi n$ is a bounded sequence as

$$|x[n]| = |\cos 0.3\pi n| \leq 1$$

- A sequence $x[n]$ is said to be **absolutely summable** if

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty$$

- Example - The following sequence is absolutely summable

$$y[n] = \begin{cases} 0.3^n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

Other Types of Classification (2/2)

- A sequence $x[n]$ is said to be **square** summable if ◦

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 < \infty$$

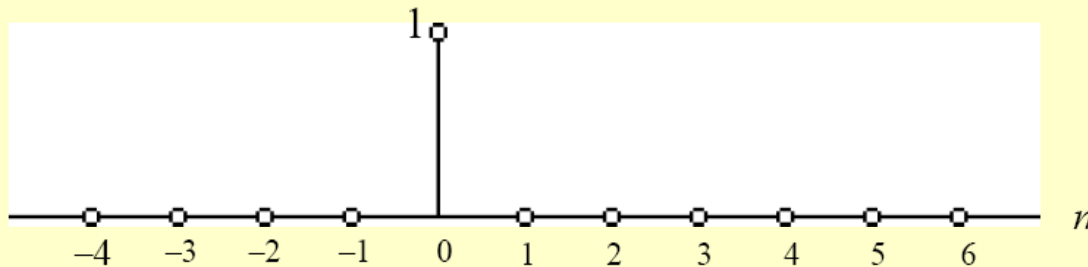
- Example - The sequence

$$h[n] = \frac{\sin 0.4n}{\pi n}$$

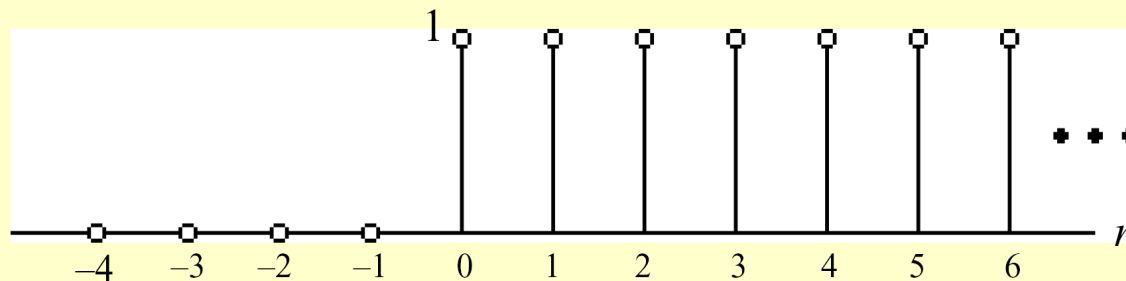
is square-summable but not absolutely summable

Basic Sequences (1/7)

- **Unit Sample Sequence** - $\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$



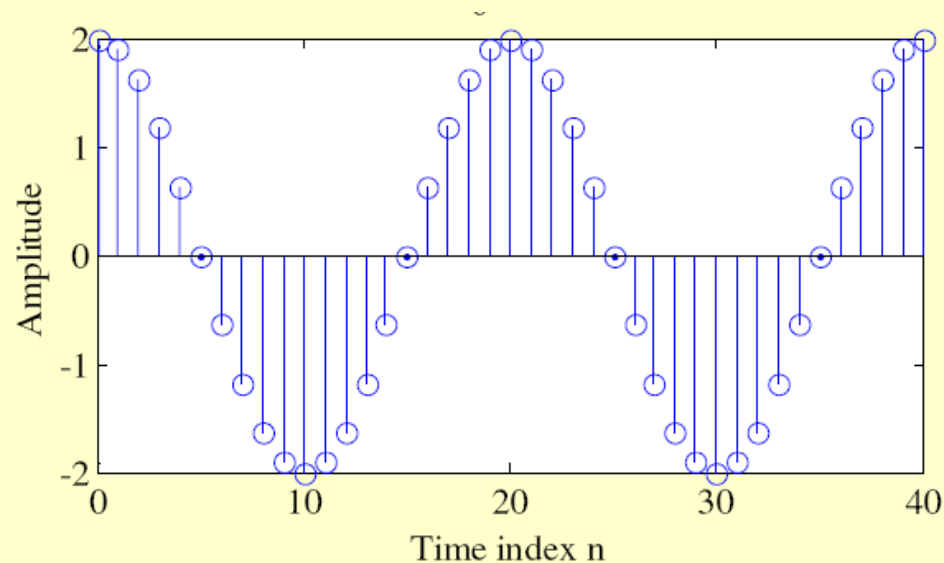
- **Unit Step Sequence** - $\mu[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$



Basic Sequences (2/7)

- **Real Sinusoidal Sequence** – $x[n] = A \cos(\omega_o n + \phi)$

Example: $A = 2$, $\omega_o = 0.1$

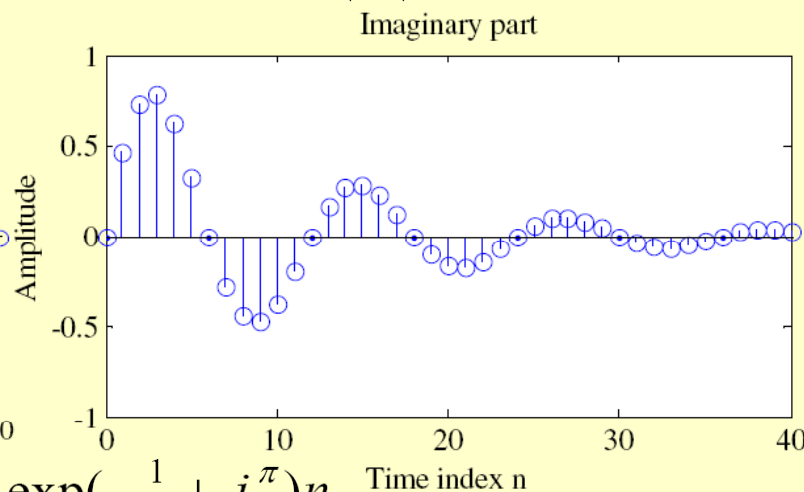
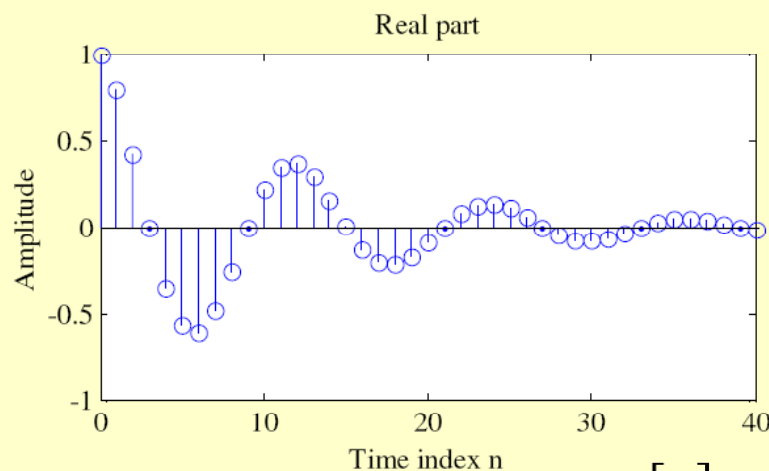


Basic Sequences (3/7)

- **Exponential Sequence** – $x[n] = A\alpha^n$, $-\infty < n < \infty$
where A and α are real or complex numbers
- If we write $\alpha = e^{(\sigma_o + j\omega_o)}$, $A = |A|e^{j\phi}$,

$$x[n] = |A|e^{j\phi}e^{(\sigma_o + j\omega_o)n} = x_{re}[n] + jx_{im}[n],$$

$$x_{re}[n] = |A|e^{\sigma_o n} \cos(\omega_o n + \phi), \quad x_{im}[n] = |A|e^{\sigma_o n} \sin(\omega_o n + \phi)$$



$$x[n] = \exp\left(-\frac{1}{12} + j\frac{\pi}{6}\right)n$$

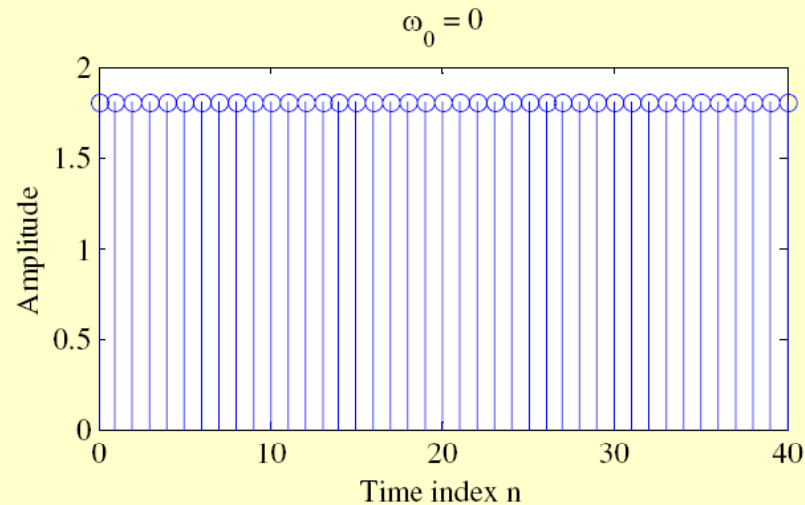
Original PowerPoint slides prepared by S. K. Mitra

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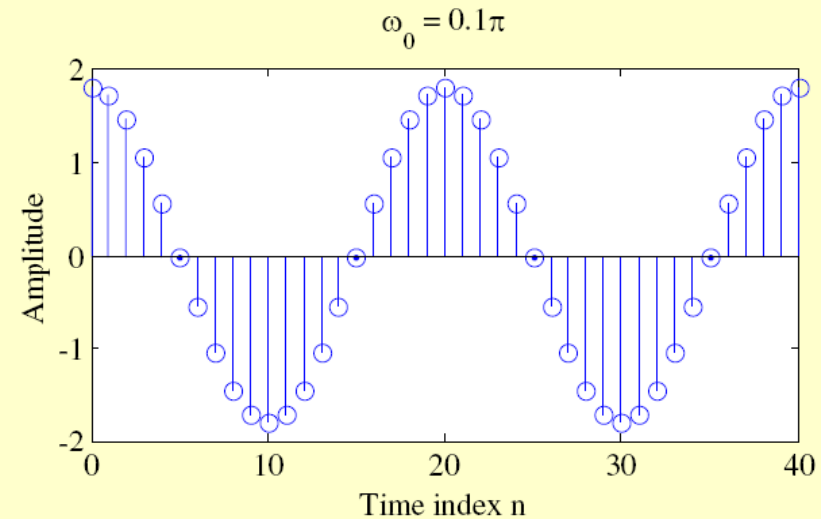
Basic Sequences (4/7)

- Sinusoidal sequence $A\cos(\omega_o n + \phi)$ and complex exponential sequence $B\exp(j\omega_o n)$ are periodic sequences of period N if $\omega_o N = 2\pi r$ where N and r are positive integers
- Smallest value of N satisfying $\omega_o N = 2\pi r$ is the **fundamental period** of the sequence
- To verify the above fact, consider $x_1[n] = \cos(\omega_o n + \phi)$
$$x_2[n] = \cos(\omega_o (n + N) + \phi)$$
$$= \cos(\omega_o n + \phi)\cos\omega_o N - \sin(\omega_o n + \phi)\sin\omega_o N$$
- $x_1[n] = x_2[n]$ if and only if $\omega_o N = 2\pi r$ or $\frac{2\pi}{\omega_o} = \frac{N}{r}$
- If $2\pi/\omega_o$ is a noninteger rational number, then the period will be a multiple of $2\pi/\omega_o$; otherwise, it's aperiodic

Basic Sequences (5/7)



- Here $\omega_0 = 0$
- period $N = \frac{2\pi r}{0} = 1$ for $r = 0$



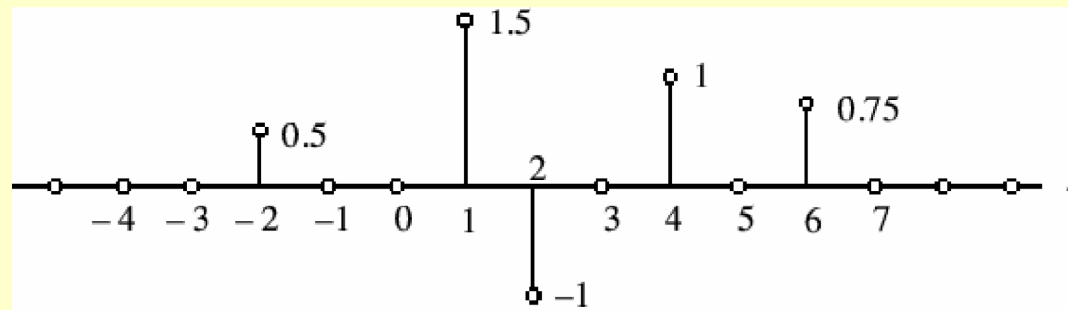
- Here $\omega_0 = 0.1\pi$
- period $N = \frac{2\pi r}{0.1\pi} = 20$ for $r = 1$

Basic Sequences (6/7)

- **Property 1** - Consider $x[n] = \exp(j\omega_1 n)$ and $y[n] = \exp(j\omega_2 n)$ with $0 \leq \omega_1 < \pi$ and $2\pi k \leq \omega_2 < 2\pi(k+1)$ where k is any positive integer
 - If $\omega_2 = \omega_1 + 2\pi k$, then $x[n] = y[n]$
 - then $x[n]$ and $y[n]$ are indistinguishable
- **Property 2** - The frequency of oscillation of $\text{Acos}(\omega_o n)$ increases as increases from 0 to π , and then decreases as increases from π to 2π
 - frequencies in the neighborhood of $\omega = 0$ (or $2k\pi$) are called **low frequencies**, whereas, frequencies in the neighborhood of $\omega = \pi$ (or $(2k+1)\pi$) are called **high frequencies**

Basic Sequences (7/7)

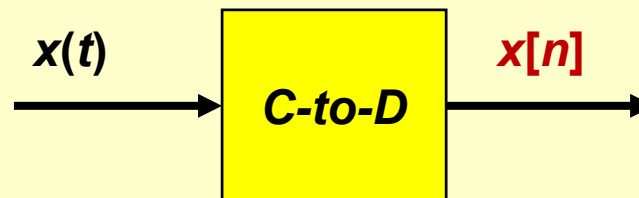
- An arbitrary sequence can be represented in the time-domain as a weighted sum of some basic sequence and its delayed (advanced) versions



$$x[n] = 0.5\delta[n+2] + 1.5\delta[n-1] - \delta[n-2] + \delta[n-4] + 0.75\delta[n-6]$$

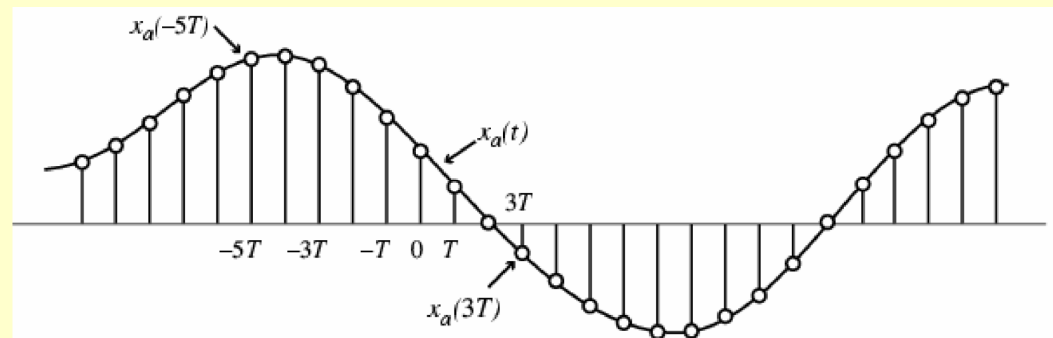
The Sampling Process (1/3)

- Sampling Process
 - Convert $x(t)$ to **numbers** $x[n]$
 - “ n ” is an integer; $x[n]$ is a sequence of values
 - Think of “ n ” as the storage address in memory
- Uniform Sampling at $t = nT$
 - IDEAL: $x[n] = x(nT)$



The Sampling Process (2/3)

- Often, a discrete-time sequence $x[n]$ is developed by uniformly sampling a continuous-time signal as follows



$$x[n] = x_a(t) \Big|_{t=nT} = x_a(nT), \quad n = \dots, -2, -1, 0, 1, 2, \dots$$

- Time variable t of $x_a(t)$ is related to the time variable n of $x[n]$ only at discrete-time instants given by

$$t_n = nT = \frac{n}{F_T} = \frac{2\pi n}{\Omega_T} \quad \begin{aligned} F_T &= 1/T \\ \Omega_T &= 2\pi F_T \end{aligned}$$

The Sampling Process (3/3)

- Consider the continuous-time signal

$$x(t) = A \cos(2\pi f_o t + \phi) = A \cos(\Omega_o t + \phi)$$

- The corresponding discrete-time signal is

$$\begin{aligned} x[n] &= A \cos(\Omega_o n T + \phi) = A \cos\left(\frac{2\pi\Omega_o}{\Omega_T} n + \phi\right) \\ &= A \cos(\omega_o n + \phi) \end{aligned}$$

where $\omega_o = 2\pi\Omega_o / \Omega_T = \Omega_o T$

is the normalized digital angular frequency of $x[n]$

- The unit of normalized digital angular frequency ω_o is radians/sample (Ω_o : radians/second)

Sampling for Audio CD

- $x[n]$ is a sampled sinusoid
 - A list of numbers stored in memory
- Example: audio CD
- CD rate is 44,100 samples per second
 - 16-bit samples
 - Stereo uses 2 channels
- Number of bytes for 1 minute is
 - $2 \times (16/8) \times 60 \times 44100 = 10.584$ Mbytes
 - So, a CD-ROM of 680-Mbyte can store up to about one-hour music
 - What about MP3?

Ambiguity in Sampling

- Sample the following three signals at 10 Hz

$$g_1(t) = \cos(6\pi t)$$

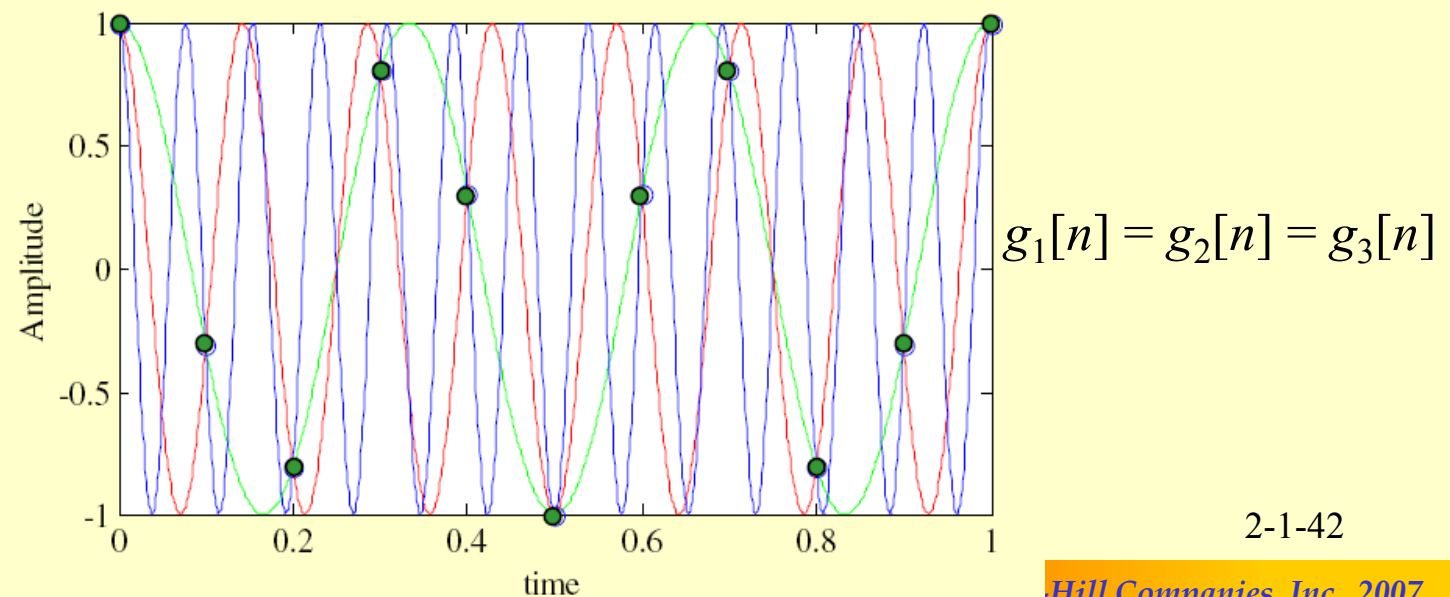
$$g_2(t) = \cos(14\pi t)$$

$$g_3(t) = \cos(26\pi t)$$

- we obtain

$$g_1[n] = \cos(0.6\pi n) \quad g_2[n] = \cos(1.4\pi n)$$

$$g_3[n] = \cos(2.6\pi n)$$



Aliasing (1/2)

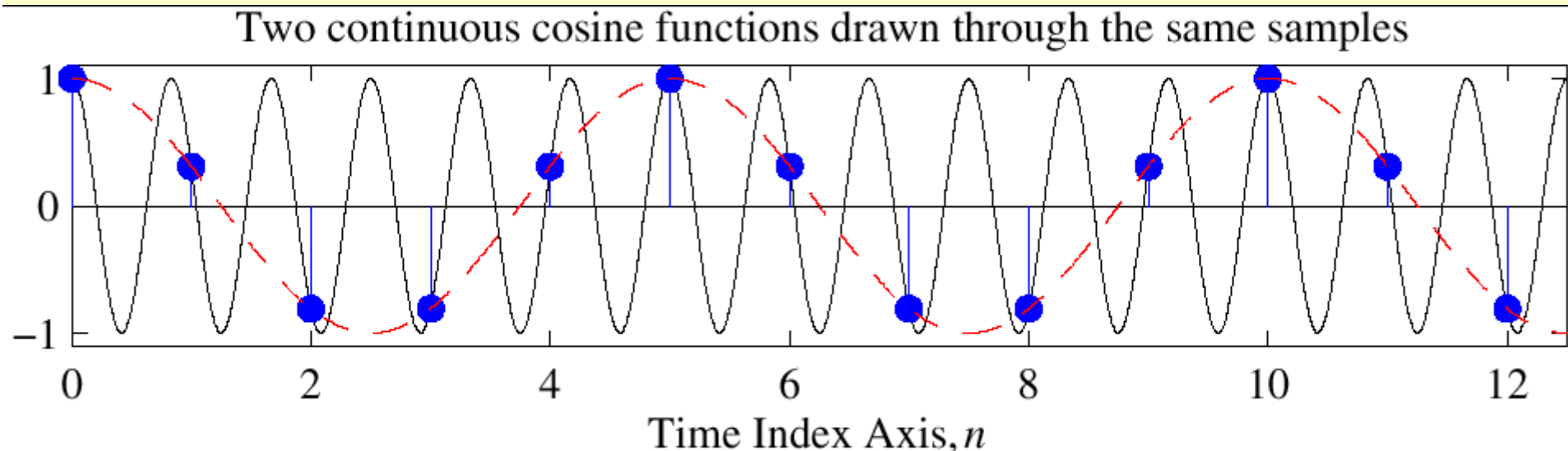
- The phenomenon of a continuous-time signal of higher frequency acquiring the identity of a sinusoidal sequence of lower frequency after sampling is called **aliasing**
- There are an infinite number of continuous-time signals that can lead to the same sequence when sampled periodically
- The family of continuous-time sinusoids leads to identical sampled signals

$$x_{a,k}(t) = A \cos((\Omega_o t + \phi) + k\Omega_T t), \quad k = 0, \pm 1, \pm 2, \dots$$

$$\begin{aligned} x_{a,k}(nT) &= A \cos((\Omega_o + k\Omega_T)nT + \phi) = A \cos\left(\frac{2\pi(\Omega_o + k\Omega_T)n}{\Omega_T} + \phi\right) \\ &= A \cos\left(\frac{2\pi\Omega_o n}{\Omega_T} + \phi\right) = a \cos(\omega_o n + \phi) = x[n] \end{aligned}$$

Aliasing (2/2)

Given the samples, draw a sinusoid through the values

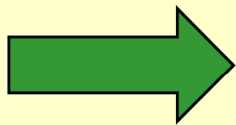


$$x[n] = \cos(0.4\pi n)$$

When n is an integer
 $\cos(0.4\pi n) = \cos(2.4\pi n)$

Sampling Theorem (1/2)

- Recall $\omega_o = \frac{2\pi\Omega_o}{\Omega_T}$
- Thus if $\Omega_T > 2\Omega_o$, then the corresponding normalized digital angular frequency ω_o of the discrete-time signal obtained by sampling the parent continuous-time sinusoidal signal will be in the range $-\pi < \omega < \pi$



No Aliasing

- On the other hand, if $\Omega_T < 2\Omega_o$, the normalized digital angular frequency will foldover into a lower digital frequency $\omega_o = \langle 2\pi\Omega_o / \Omega_T \rangle_{2\pi}$ in the range $-\pi < \omega < \pi$ because of aliasing

Sampling Theorem (2/2)

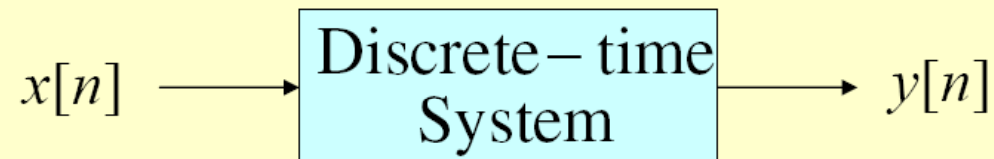
- To prevent aliasing, the sampling frequency Ω_T should be greater than 2 times the frequency Ω_o of the sinusoidal signal being sampled
- Generalization: Consider an arbitrary continuous-time signal $x(t)$ composed of a weighted sum of a number of sinusoidal signals

Shannon Sampling Theorem

A continuous-time signal $x(t)$ with frequencies no higher than $f_{\max}$ can be reconstructed exactly from its samples $x[n] = x(nT_s)$, if the samples are taken at a rate $f_s = 1/T_s$ that is greater than $2f_{\max}$.

Discrete-Time System

- A discrete-time system processes a given input sequence $x[n]$ to generate an output sequence $y[n]$ with more desirable properties
- In most applications, the discrete-time system is a single-input, single-output system:



- 2-input, 1-output discrete-time systems - Modulator, adder
- 1-input, 1-output discrete-time systems - Multiplier, unit delay, unit advance

Discrete-Time System Examples (1/10)

- **Accumulator** -
$$y[n] = \sum_{\ell=-\infty}^n x[\ell]$$
$$= \sum_{\ell=-\infty}^{n-1} x[\ell] + x[n] = y[n-1] + x[n]$$

- The output $y[n]$ is the sum of the input sample $x[n]$ and the previous output $y[n-1]$
- The system cumulatively adds, i.e., it accumulates all input sample values
- Input-output relation can also be written in the form

$$y[n] = \sum_{\ell=-\infty}^{-1} x[\ell] + \sum_{\ell=0}^n x[\ell] = y[-1] + \sum_{\ell=0}^n x[\ell], \quad n \geq 0$$

- The second form is used for a causal input sequence, in which case $y[-1]$ is called the **initial condition**

Discrete-Time System Examples (2/10)

- **M-point moving-average system** -

$$y[n] = \frac{1}{M} \sum_{k=0}^{M-1} x[n-k]$$

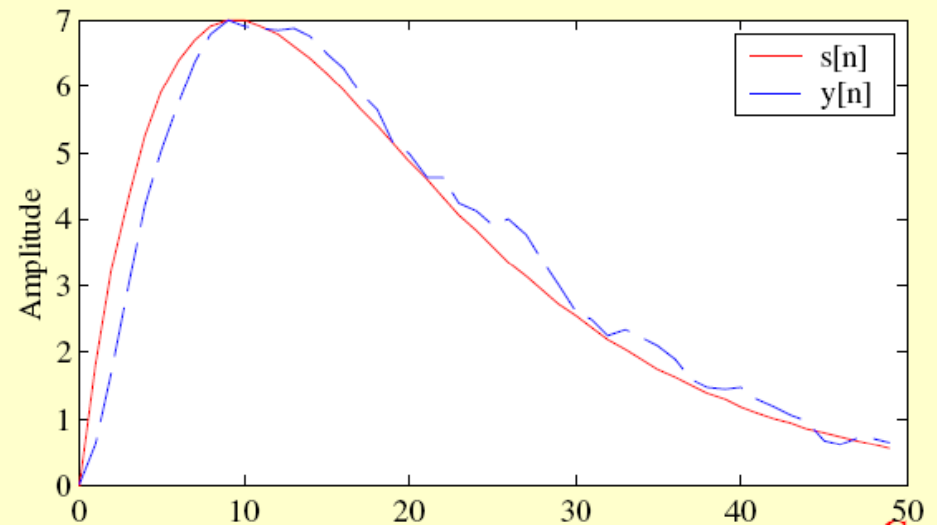
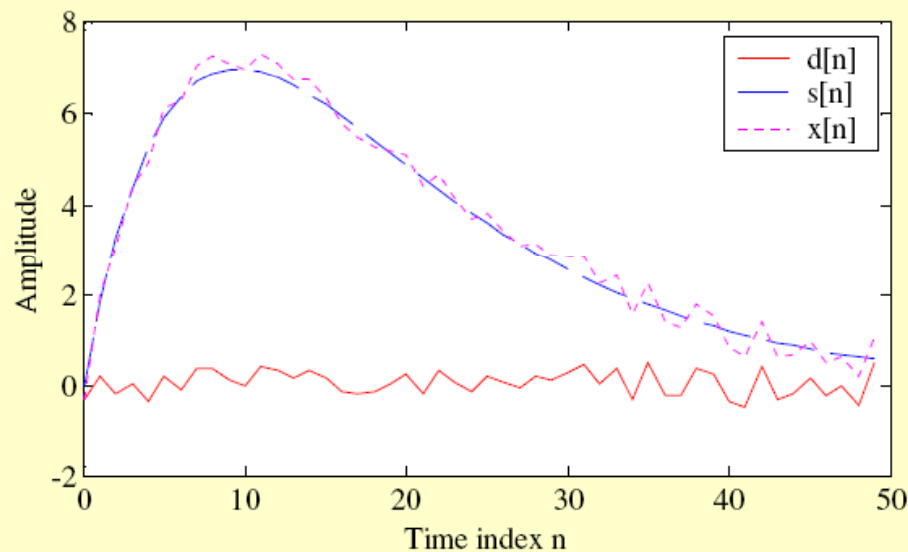
- Used in smoothing random variations in data
- A direct implementation of the *M-point* moving average system requires $M - 1$ additions, 1 division
- A more efficient implementation, which requires 2 additions and 1 division

$$\begin{aligned} y[n] &= \frac{1}{M} \left(\sum_{\ell=0}^{M-1} x[n-\ell] + x[n-M] - x[n-M] \right) \\ &= y[n-1] + \frac{1}{M} (x[n] - x[n-M]) \end{aligned}$$

Discrete-Time System Examples (3/10)

- An application: Consider $x[n] = s[n] + d[n]$ where $s[n]$ is the signal corrupted by a noise $d[n]$

$$s[n] = 2[n(0.9)^n], \quad d[n] - \text{random signal}$$



Discrete-Time System Examples (4/10)

- **Exponentially Weighted Running Average Filter**

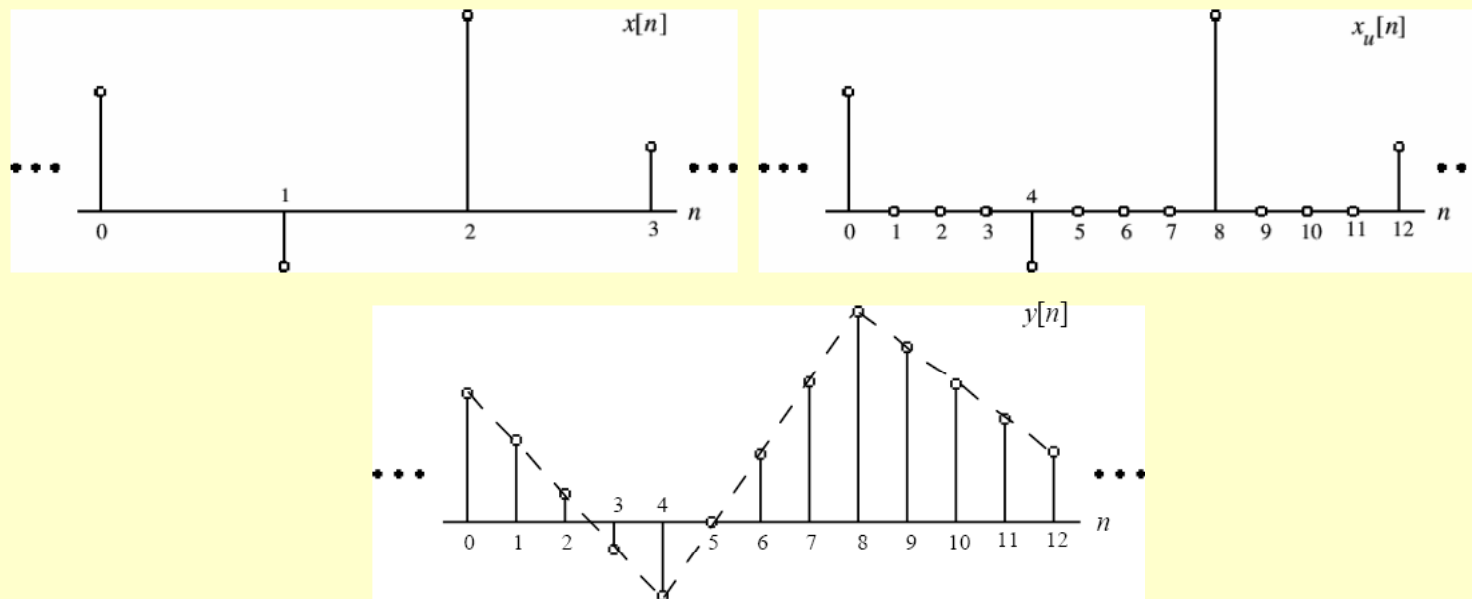
$$y[n] = \alpha y[n-1] + x[n],$$

- requires only 2 additions, 1 multiplication and storage of the previous running average
- does not require storage of past input data samples
- the filter places more emphasis on current data samples and less emphasis on past ones as illustrated below

$$\begin{aligned} y[n] &= \alpha(\alpha y[n-2] + x[n-1]) + x[n] \\ &= \alpha^2 y[n-2] + \alpha x[n-1] + x[n] \\ &= \alpha^2(\alpha y[n-3] + x[n-2]) + \alpha x[n-1] + x[n] \\ &= \alpha^3 y[n-3] + \alpha^2 x[n-2] + \alpha x[n-1] + x[n] \end{aligned} \quad 2-2-51$$

Discrete-Time System Examples (5/10)

- **Linear interpolation** - employed to estimate sample values between pairs of adjacent sample values of a discrete-time sequence
- **Factor-of-4 interpolation**



Discrete-Time System Examples (6/10)

- **Factor-of-2 interpolator**

$$y[n] = x_u[n] + \frac{1}{2}(x_u[n-1] + x_u[n+1])$$

- **Factor-of-3 interpolator**

$$y[n] = x_u[n] + \frac{1}{3}(x_u[n-1] + x_u[n+2]) \\ + \frac{2}{3}(x_u[n-2] + x_u[n+1])$$

Discrete-Time System Examples (7/10)

- Factor-of-2 interpolator



Original (512×512)



Down-sampled
(256×256)



Interpolated (512 × 512)

Discrete-Time System Examples (8/10)

Median Filter –

- The median of a set of $(2K+1)$ numbers *is* the number such that K numbers from the set have values greater than this number and the other K numbers have values smaller
- Median can be determined by rank-ordering the numbers in the set by their values and choosing the number at the middle
- Example: Consider the set of numbers
$$\{2, -3, 10, 5, -1\}$$
- Rank-order set is given by
$$\{-3, -1, 2, 5, 10\}$$
- $\text{med}\{2, -3, 10, 5, -1\} = 2$

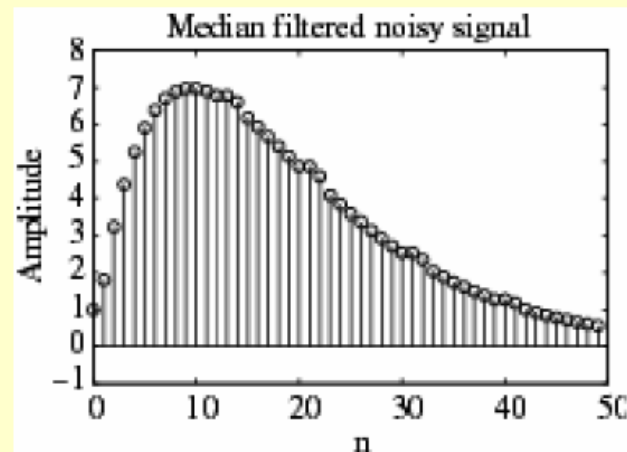
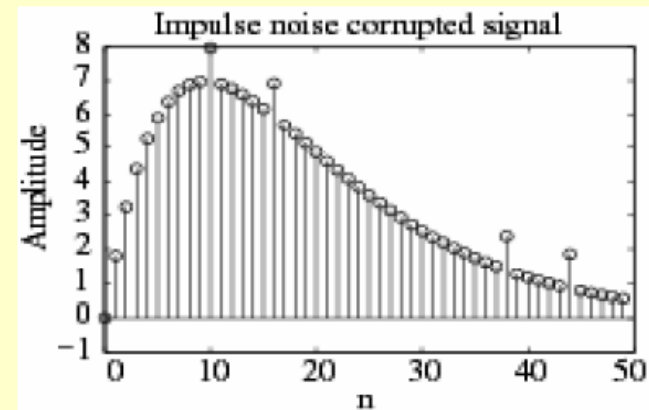
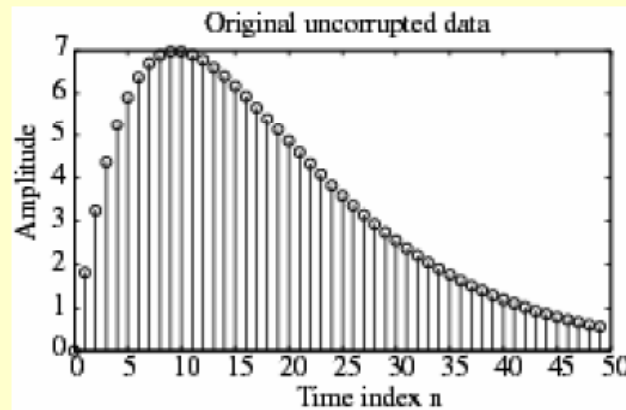
Discrete-Time System Examples (9/10)

Median Filter –

- Implemented by sliding a window of odd length over the input sequence $\{x[n]\}$ one sample at a time
- Output $y[n]$ at instant n is the median value of the samples inside the window centered at n
- Useful in removing additive random noise, which shows up as sudden large errors in the corrupted signal
- Usually used for the smoothing of signals corrupted by impulse noise

Discrete-Time System Examples (10/10)

- **Median Filtering Example –**



Classifications of Discrete-Time Systems

- Linear system
- Shift-invariant system)
- Causal system
- Stable system
- Passive and lossless system

Linear Discrete-Time Systems (1/3)

- **Definition** - If $y_1[n]$ is the output due to an input $x_1[n]$ and $y_2[n]$ is the output due to an input $x_2[n]$ then for an input

$$x[n] = \alpha x_1[n] + \beta x_2[n]$$

the output is given by

$$y[n] = \alpha y_1[n] + \beta y_2[n]$$

for any arbitrary constants α and β

- **Example: Accumulator** – $y_1[n] = \sum_{\ell=-\infty}^n x_1[\ell]$, $y_2[n] = \sum_{\ell=-\infty}^n x_2[\ell]$

For an input

$$x[n] = \alpha x_1[n] + \beta x_2[n]$$

the output is $y[n] = \sum_{\ell=-\infty}^n (\alpha x_1[\ell] + \beta x_2[\ell])$

$$= \alpha \sum_{\ell=-\infty}^n x_1[\ell] + \beta \sum_{\ell=-\infty}^n x_2[\ell] = \alpha y_1[n] + \beta y_2[n]$$

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Linear Discrete-Time Systems (2/3)

- If the outputs $y_1[n]$ and $y_2[n]$ for inputs $x_1[n]$ and $x_2[n]$ are given by
$$y_1[n] = y_1[-1] + \sum_{l=0}^n x_1[l]$$

$$y_2[n] = y_2[-1] + \sum_{l=0}^n x_2[l]$$

- The output $y[n]$ for an input $\alpha x_1[n] + \beta x_2[n]$ is

$$y[n] = y[-1] + \sum_{\ell=0}^n (\alpha x_1[\ell] + \beta x_2[\ell])$$

- Now $\alpha y_1[n] + \beta y_2[n]$

$$= \alpha(y_1[-1] + \sum_{l=0}^n x_1[l]) + \beta(y_2[-1] + \sum_{l=0}^n x_2[l])$$

$$= (\alpha y_1[-1] + \beta y_2[-1]) + (\alpha \sum_{l=0}^n x_1[l] + \beta \sum_{l=0}^n x_2[l])$$

Linear Discrete-Time Systems (3/3)

- For the causal accumulator to be **linear** the condition $y[-1] = \alpha y_1[-1] + \beta y_2[-1]$ must hold for all initial conditions $y[-1]$, $y_1[-1]$, $y_2[-1]$, and constants α and β
- This condition cannot be satisfied unless the accumulator is initially at rest with zero initial condition
- For nonzero initial condition, the system is **nonlinear**

Non-Linear Discrete-Time Systems

- The median filter described earlier is a nonlinear discrete-time system
- To show this, consider a median filter with a window of length 3
- The output of the filter for an input $\{x_1[n]\} = \{3, 4, 5\}$, $0 \leq n \leq 2$, is $\{y_1[n]\} = \{3, 4, 4\}$, $0 \leq n \leq 2$
- The output for an input $\{x_2[n]\} = \{2, -1, -1\}$, $0 \leq n \leq 2$ is $\{y_2[n]\} = \{0, -1, -1\}$, $0 \leq n \leq 2$
- However, the output for an input $\{x[n]\} = \{x_1[n] + x_2[n]\}$ is $\{y[n]\} = \{3, 4, 3\} \neq \{y_1[n] + y_2[n]\} = \{3, 3, 3\}$
- Hence, the median filter is a nonlinear discrete-time system

Shift Invariant Systems (1/2)

- For a shift-invariant system, if $y_1[n]$ is the response to an input $x_1[n]$, then the response to an input $x[n] = x_1[n - n_0]$ is simply

$$y[n] = y_1[n - n_0]$$

where n_0 is any positive or negative integer

- In the case of sequences and systems with indices n related to discrete instants of time, the above property is called **time-invariance** property
- Time-invariance property ensures that for a specified input, the output is independent of the time the input is being applied

Shift Invariant Systems (2/2)

- **Example** – Consider the following up-sampler

$$x_u[n] = \begin{cases} x[n/L], & n = 0, \pm L, \pm 2L, \dots \\ 0, & \text{otherwise} \end{cases}$$

- for input $x_1[n] = x[n - n_o]$, the output $x_{1,u}[n]$ is

$$\begin{aligned} x_{1,u}[n] &= \begin{cases} x_1[n/L], & n = 0, \pm L, \pm 2L, \dots \\ 0, & \text{otherwise} \end{cases} \\ &= \begin{cases} x[(n - Ln_o)/L], & n = 0, \pm L, \pm 2L, \dots \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$

- However, the upsampler is a time-varying system because

$$\begin{aligned} x_u[n - n_o] &= \begin{cases} x[(n - n_o)/L], & n = n_o, n_o \pm L, n_o \pm 2L, \dots \\ 0, & \text{otherwise} \end{cases} \\ &\neq x_{1,u}[n] \end{aligned}$$

Linear Time-Invariant Systems

- **Linear Time-Invariant (LTI) System** - A system satisfying both the linearity and the time-invariance property
- LTI systems are mathematically easy to analyze and characterize, and consequently, easy to design
- Highly useful signal processing algorithms have been developed utilizing this class of systems over the last several decades

Causal Systems (1/2)

- In a **causal system**, the n_o -th output sample $y[n_o]$ depends only on input samples $x[n]$ for $n \leq n_o$ and does not depend on input samples for $n > n_o$
- Let $y_1[n]$ and $y_2[n]$ be the responses of a causal system to the inputs $x_1[n]$ and $x_2[n]$, respectively. Then

$$x_1[n] = x_2[n] \text{ for } n < N$$

Implied also that

$$y_1[n] = y_2[n] \text{ for } n < N$$

- For a causal system, changes in output samples do not precede changes in the input samples

Causal Systems (2/2)

- Examples of causal systems:

$$y[n] = \alpha_1 x[n] + \alpha_2 x[n-1] + \alpha_3 x[n-2] + \alpha_4 x[n-3]$$

$$y[n] = b_0 x[n] + b_1 x[n-1] + b_2 x[n-2] \\ + a_1 y[n-1] + a_2 y[n-2]$$

$$y[n] = y[n-1] + x[n]$$

- Examples of noncausal systems:

$$y[n] = x_u[n] + 1/2 (x_u[n-1] + x_u[n+1])$$

$$y[n] = x_u[n] + 1/3 (x_u[n-1] + x_u[n+2]) + 2/3 (x_u[n-2] + x_u[n+1])$$

- A noncausal system can be made causal by delaying the output by an appropriate number of samples
- A causal implementation of the factor-of-2 interpolator

$$y[n] = x_u[n-1] + \frac{1}{2}(x_u[n-2] + x_u[n])$$

Stable Systems

- **Bounded-Input, Bounded Output (BIBO)** stability
- If $y[n]$ is the response to an input $x[n]$ and if

then

$$|x[n]| \leq B_x \quad \text{for all values of } n$$
$$|y[n]| \leq B_y \quad \text{for all values of } n$$

- Example – the M -point moving average filter is BIBO stable

$$y[n] = \frac{1}{M} \sum_{k=0}^{M-1} x[n-k]$$

- With a bounded input $|x[n]| \leq B_x$

$$|y[n]| = \left| \frac{1}{M} \sum_{k=0}^{M-1} x[n-k] \right| \leq \frac{1}{M} \sum_{k=0}^{M-1} |x[n-k]|$$
$$\leq \frac{1}{M} (MB_x) \leq B_x$$

Passive and Lossless Systems

- A discrete-time system is defined to be **passive** if, for every finite-energy input $x[n]$, the output $y[n]$ has, at most, the same energy

$$\sum_{n=-\infty}^{\infty} |y[n]|^2 \leq \sum_{n=-\infty}^{\infty} |x[n]|^2$$

- For a **lossless system**, the above inequality is satisfied with an equal sign for every input
- Example - Consider the discrete-time system defined by $y[n] = \alpha x[n - N]$ with N a positive integer
- Its output energy is given by

$$\sum_{n=-\infty}^{\infty} |y[n]|^2 = |\alpha|^2 \sum_{n=-\infty}^{\infty} |x[n]|^2$$

passive system if $|\alpha| < 1$, and lossless if $|\alpha| = 1$

Impulse & Step Responses

- The response of a discrete-time system to a unit sample sequence $\{\delta[n]\}$ is called, the **(unit) impulse response**, and is denoted by $\{h[n]\}$
- The response of a discrete-time system to a unit step sequence $\{\mu[n]\}$, is called the **(unit) step response**, and is denoted by $\{s[n]\}$
- Example - The impulse response $\{h[n]\}$ of the factor-of-2 interpolator

$$y[n] = \alpha_1 x[n] + \alpha_2 x[n-1] + \alpha_3 x[n-2] + \alpha_4 x[n-3]$$

Is obtained by

$$h[n] = \alpha_1 \delta[n] + \alpha_2 \delta[n-1] + \alpha_3 \delta[n-2] + \alpha_4 \delta[n-3]$$

$$\{h[n]\} = \{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$$

Impulse Response

- Example - The impulse response of the discrete-time accumulator

$$y[n] = \sum_{\ell=-\infty}^n x[\ell]$$

is obtained by setting $x[n] = \delta[n]$ resulting in

$$h[n] = \sum_{\ell=-\infty}^n \delta[\ell] = \mu[n]$$

- Example - The impulse response $\{h[n]\}$ of the factor-of-2 interpolator

$$y[n] = x_u[n] + \frac{1}{2}(x_u[n-1] + x_u[n+1])$$

is

$$h[n] = \delta[n] + \frac{1}{2}(\delta[n-1] + \delta[n+1])$$

$$\{h[n]\} = \{0.5, \underset{\uparrow}{1}, 0.5\}$$

Time-Domain Characterization of LTI Discrete-Time System (1/3)

- **Input-Output Relationship -**

A consequence of the linear, time-invariance property is that an LTI discrete-time system is completely characterized by its impulse response

➡ Knowing the impulse response, one can compute the output of the system for any arbitrary input

- Let $h[n]$ denote the impulse response of an LTI discrete-time system, we can compute $y[n]$ for the input:

$$x[n] = 0.5\delta[n+2] + 1.5\delta[n-1] - \delta[n-2] + 0.75\delta[n-5]$$

- We can compute its outputs for each member of the input separately and add the individual outputs to determine $y[n]$
(Linearity)

Time-Domain Characterization of LTI Discrete-Time System (2/3)

- Since the system is LTI

$$\begin{array}{lcl} \text{input} & & \text{output} \\ 0.5\delta[n+2] & \rightarrow & 0.5h[n+2] \\ 1.5\delta[n-1] & \rightarrow & 1.5h[n-1] \\ -\delta[n-2] & \rightarrow & -h[n-2] \\ 0.75\delta[n-5] & \rightarrow & 0.75h[n-5] \end{array}$$

- Because of the linearity property we get

$$\begin{aligned} y[n] = & 0.5h[n+2] + 1.5h[n-1] \\ & - h[n-2] + 0.75h[n-5] \end{aligned}$$

Time-Domain Characterization of LTI Discrete-Time System (3/3)

- Now, any arbitrary input sequence $x[n]$ can be expressed as a linear combination of delayed and advanced unit sample sequences in the form:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k]$$

- The response of the LTI system to an input $x[k] \delta[n - k]$ will be $x[k] h[n - k]$
- Hence, the response $y[n]$ to an input

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k]$$

will be

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k] \quad \text{or} \quad y[n] = \sum_{k=-\infty}^{\infty} x[n - k] h[k]$$

Convolution Sum (1/3)

- The summation

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{k=-\infty}^{\infty} x[n-k]h[k]$$

is called the **convolution sum** of the sequences $x[n]$ and $h[n]$ and represented as

$$y[n] = x[n] \circledast h[n]$$

- **Properties of convolution**

- Commutative property:

$$x[n] \circledast h[n] = h[n] \circledast x[n]$$

- Associative property :

$$(x[n] \circledast h[n]) \circledast y[n] = x[n] \circledast (h[n] \circledast y[n])$$

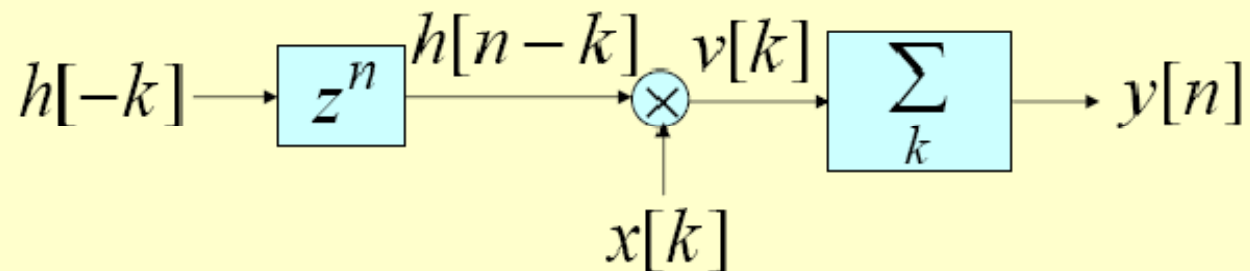
- Distributive property :

$$x[n] \circledast (h[n] + y[n]) = x[n] \circledast h[n] + x[n] \circledast y[n]$$

Convolution Sum (2/3)

- **Interpretation –**

- Time-reverse $h[k]$ to form $h[-k]$
- Shift $h[-k]$ to the right by n sampling periods if $n > 0$ (or shift to the left by n sampling periods if $n < 0$) to form $h[n-k]$
- Form the product $v[k] = x[k]h[n-k]$
- Sum all samples of $v[k]$ to develop the n -th sample of $y[n]$ of the convolution sum



Convolution Sum (3/3)

- The computation of an output sample using the convolution sum is simply a sum of products
- Involves fairly simple operations such as additions, multiplications, and delays
- We illustrate the convolution operation for the following two sequences:

$$x[n] = \begin{cases} 1, & 0 \leq n \leq 5 \\ 0, & \text{otherwise} \end{cases}$$
$$h[n] = \begin{cases} 1.8 - 0.3n, & 0 \leq n \leq 5 \\ 0, & \text{otherwise} \end{cases}$$

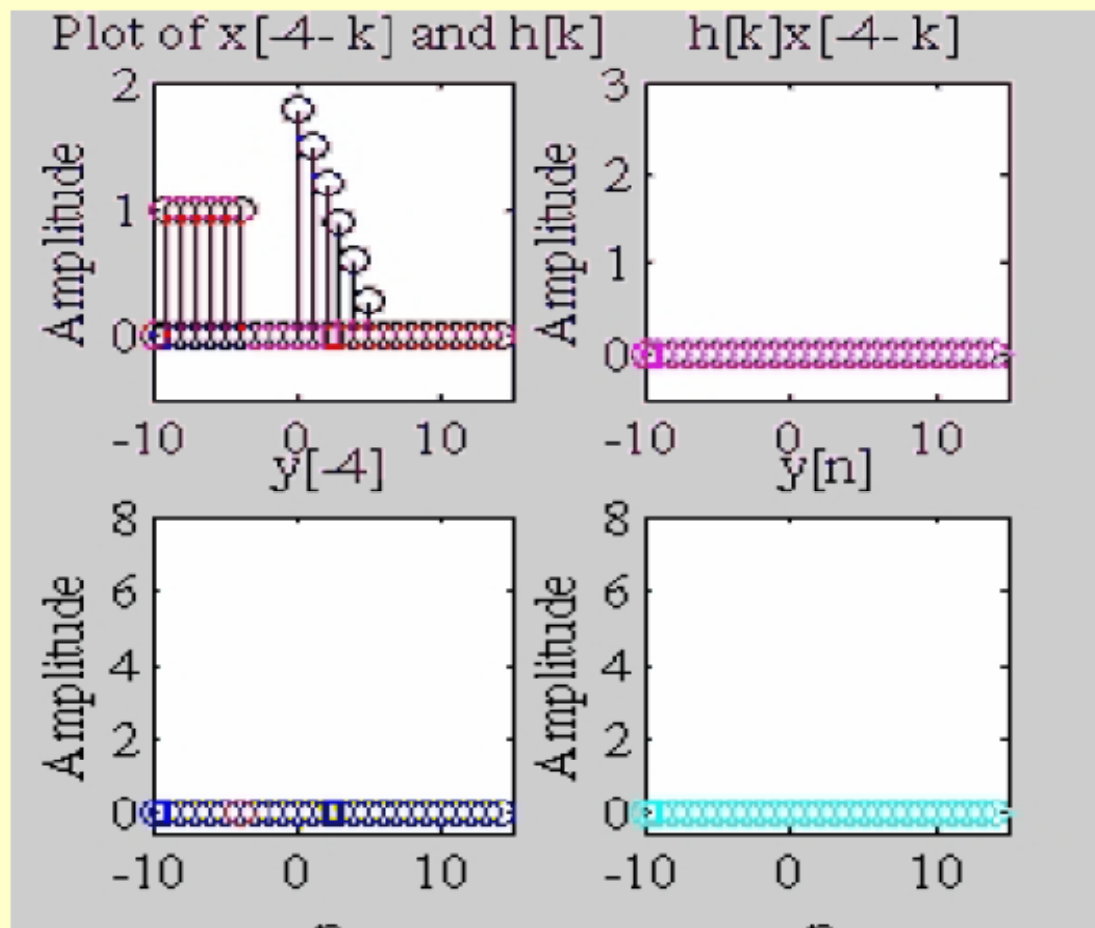
- The next several slides illustrate the convolution process

Original PowerPoint slides prepared by S. K. I.

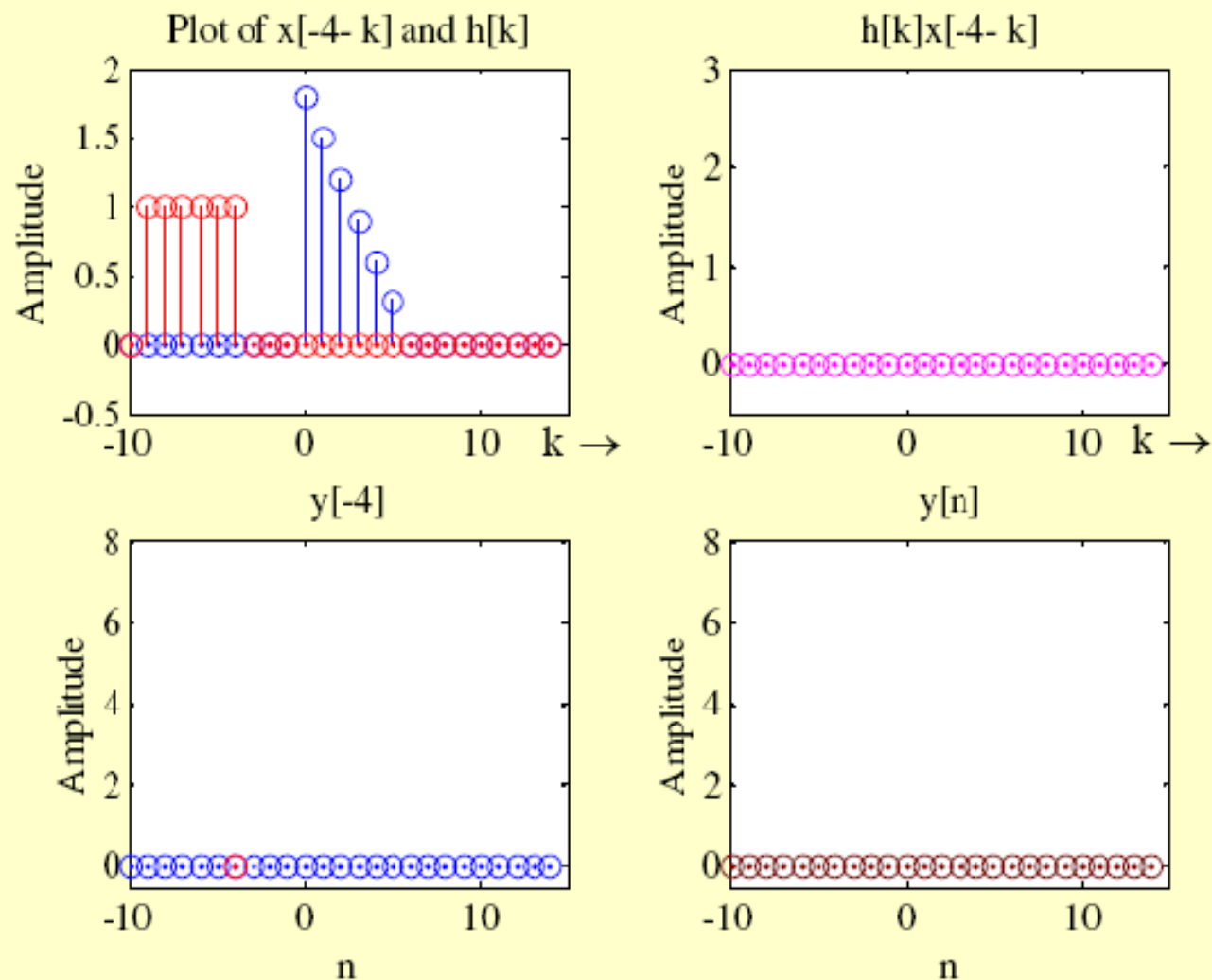
$$y[n] = x[n] \otimes h[n]$$

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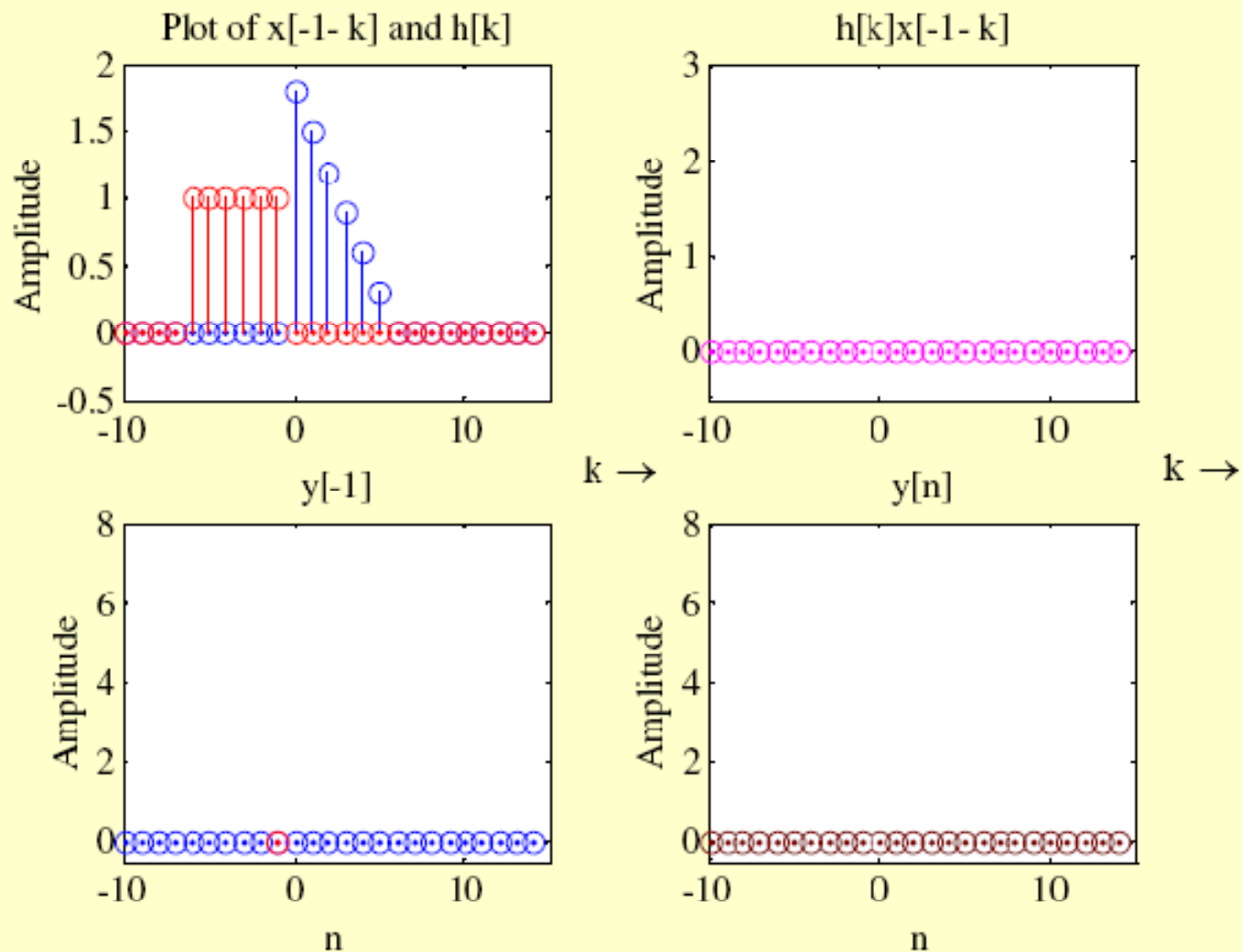
Convolution (1/12)



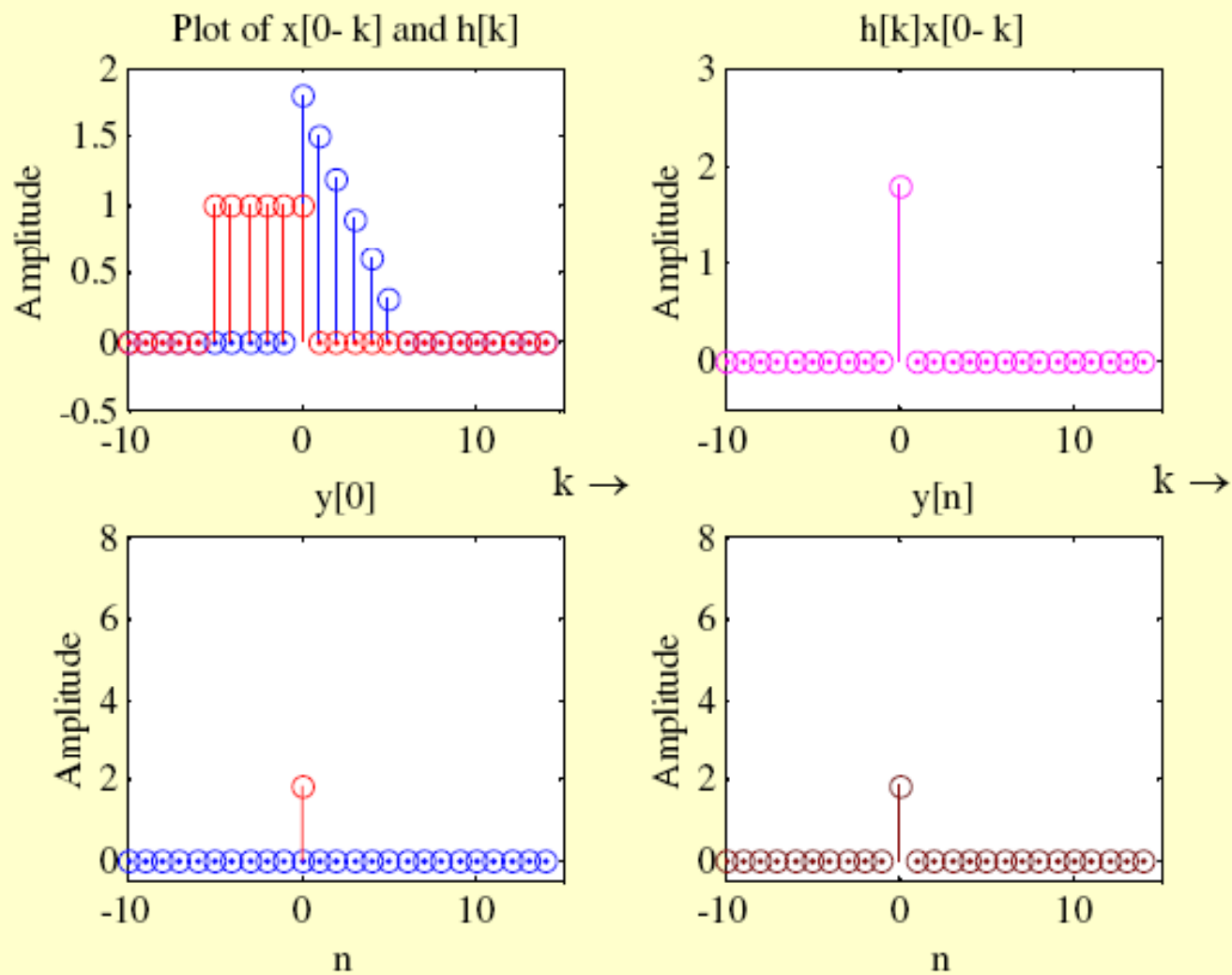
Convolution (2/12)



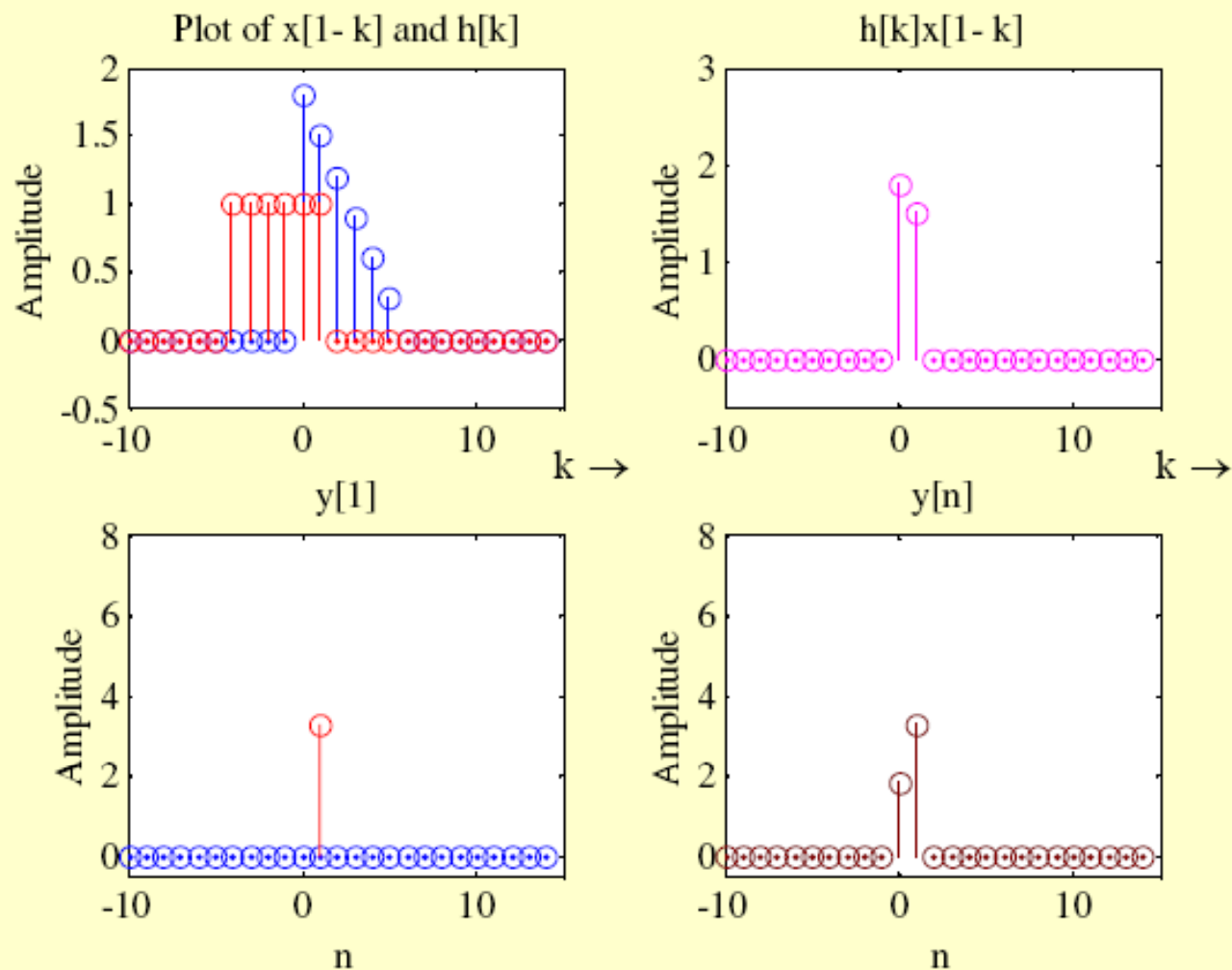
Convolution (3/12)



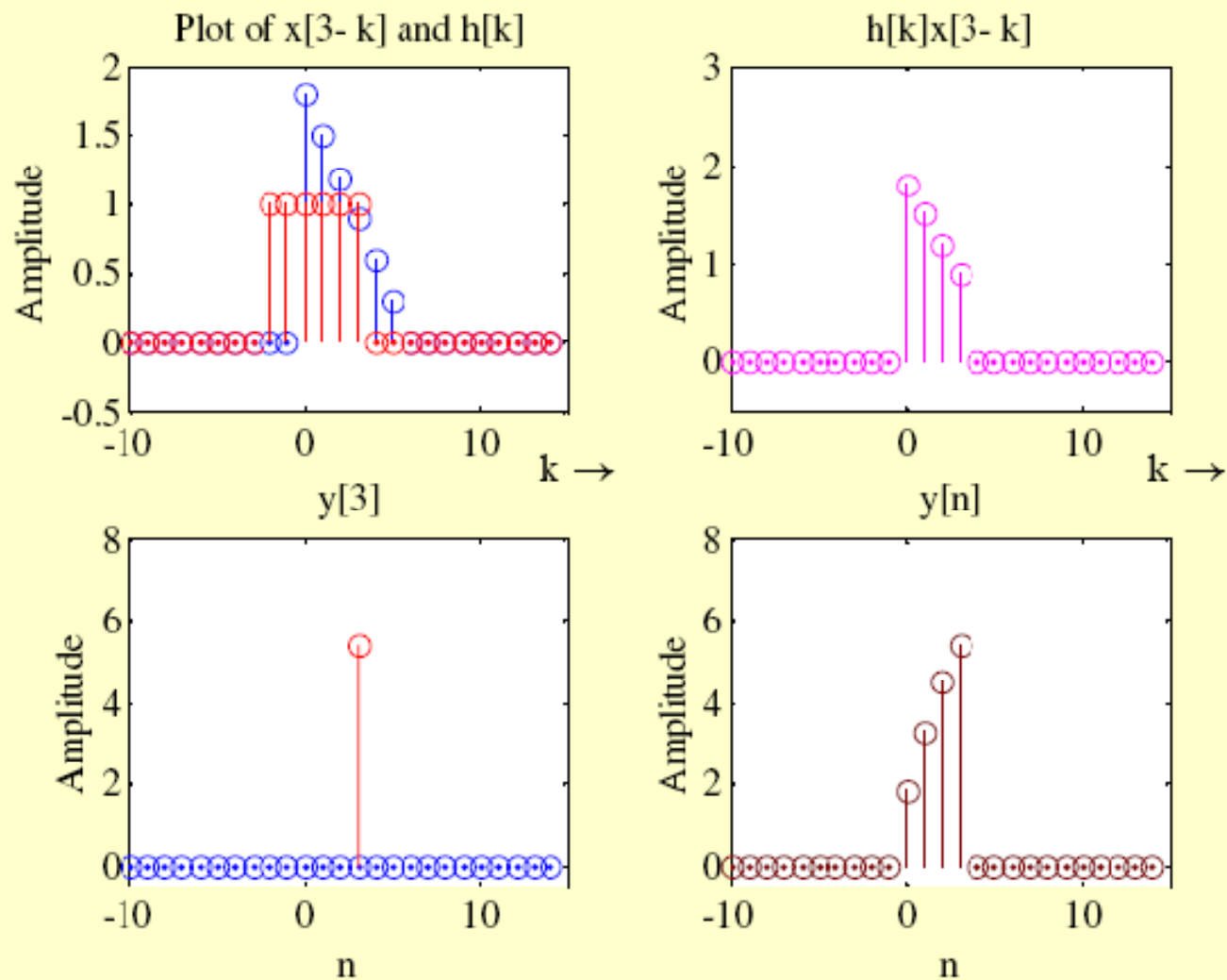
Convolution (4/12)



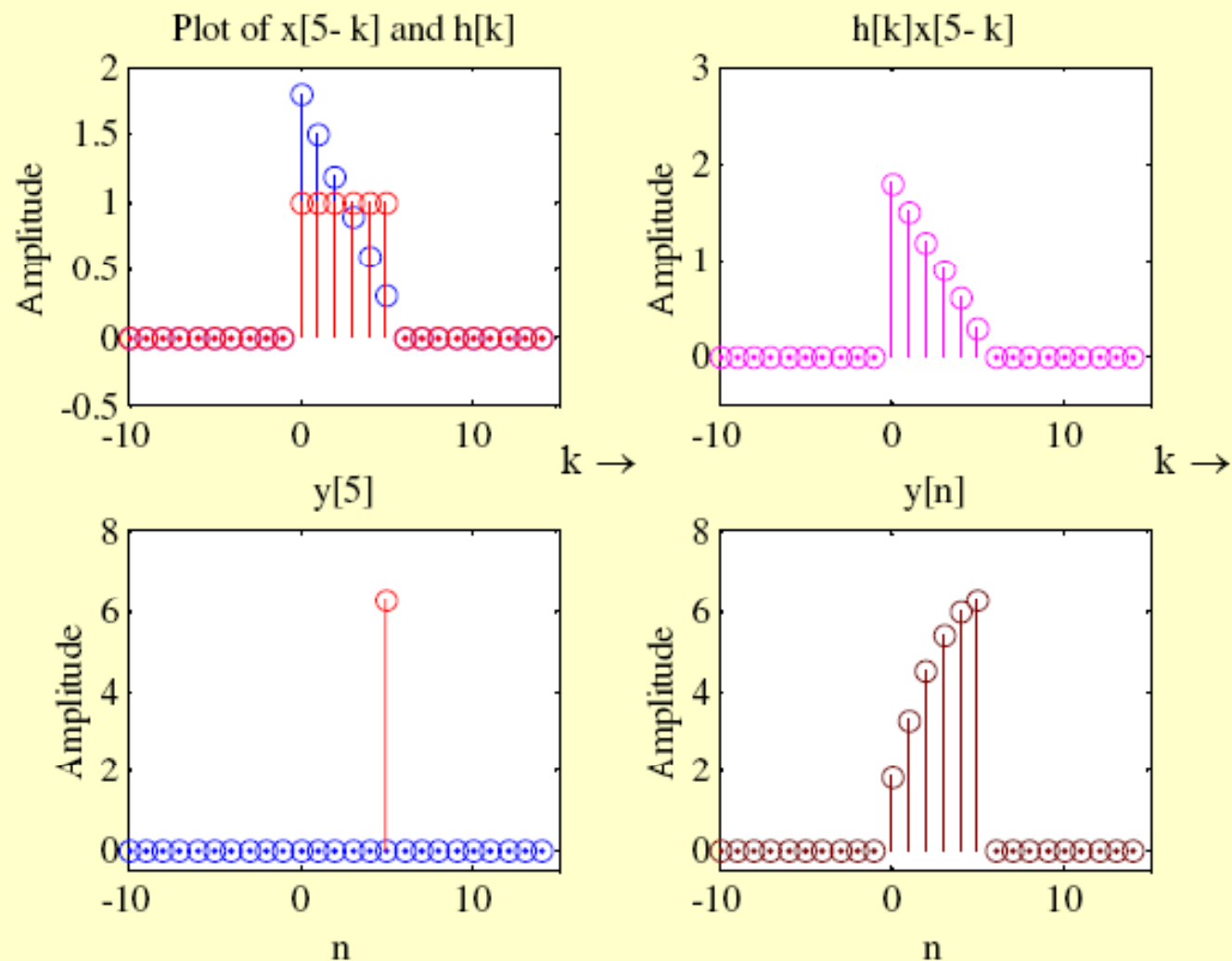
Convolution (5/12)



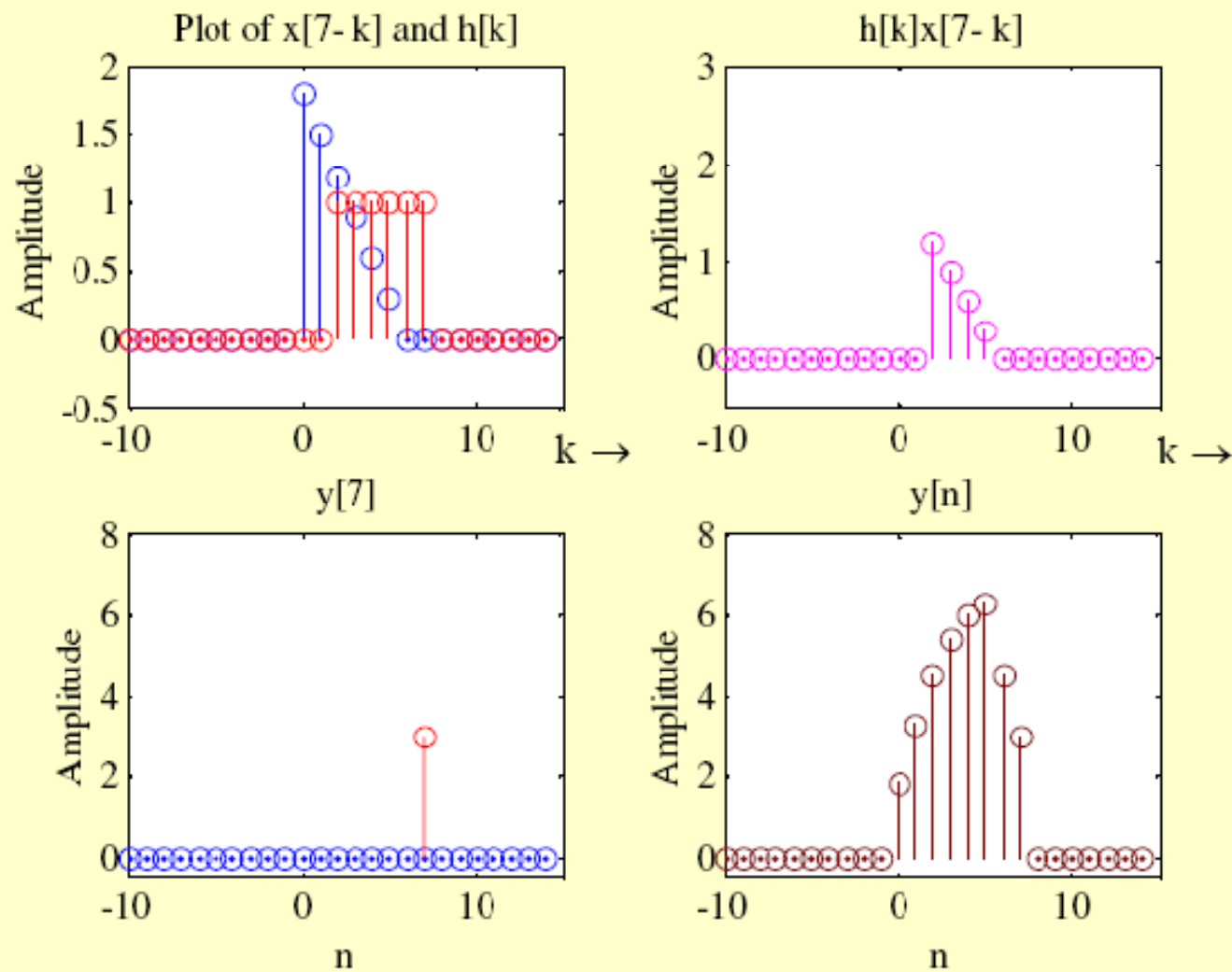
Convolution (6/12)



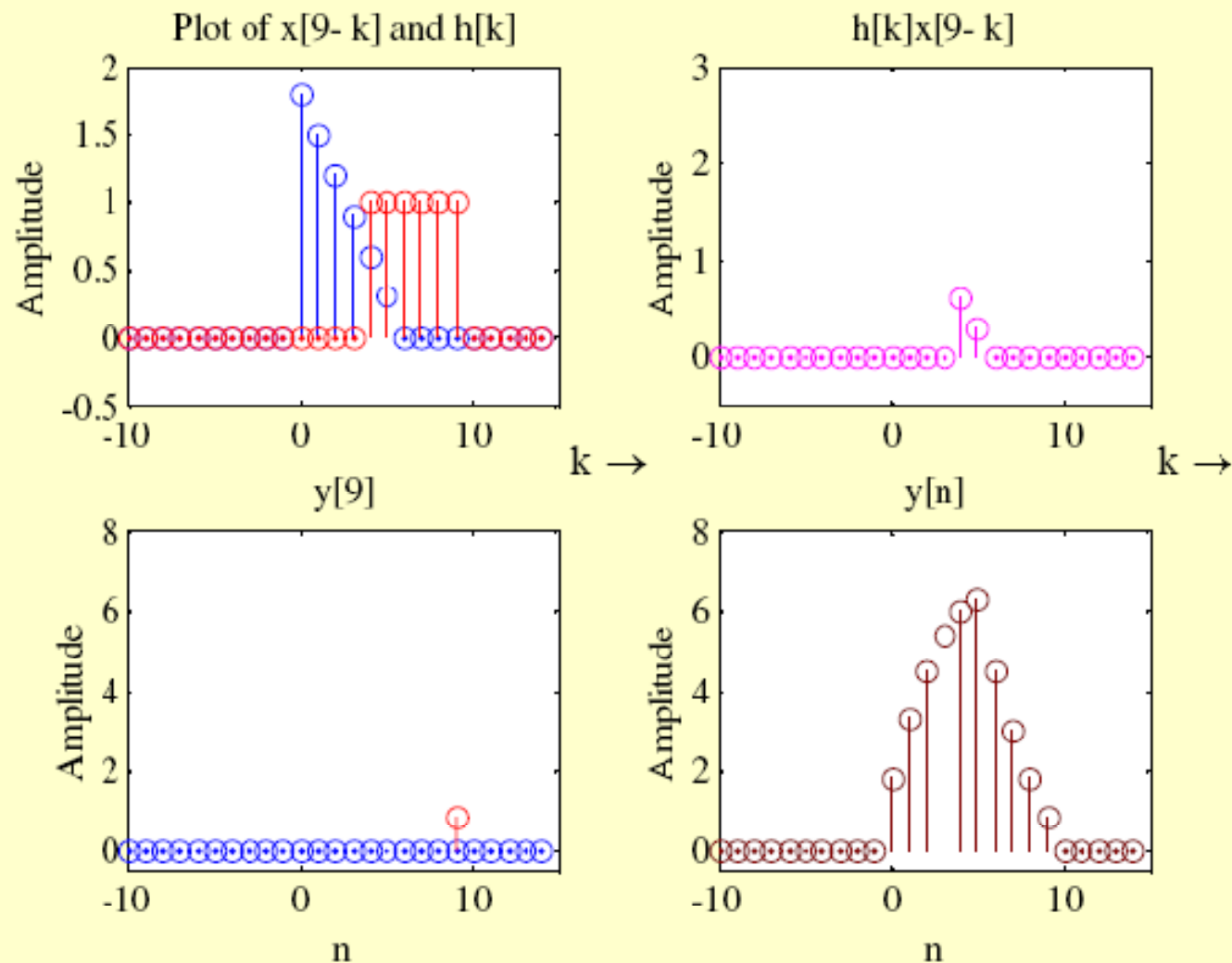
Convolution (7/12)



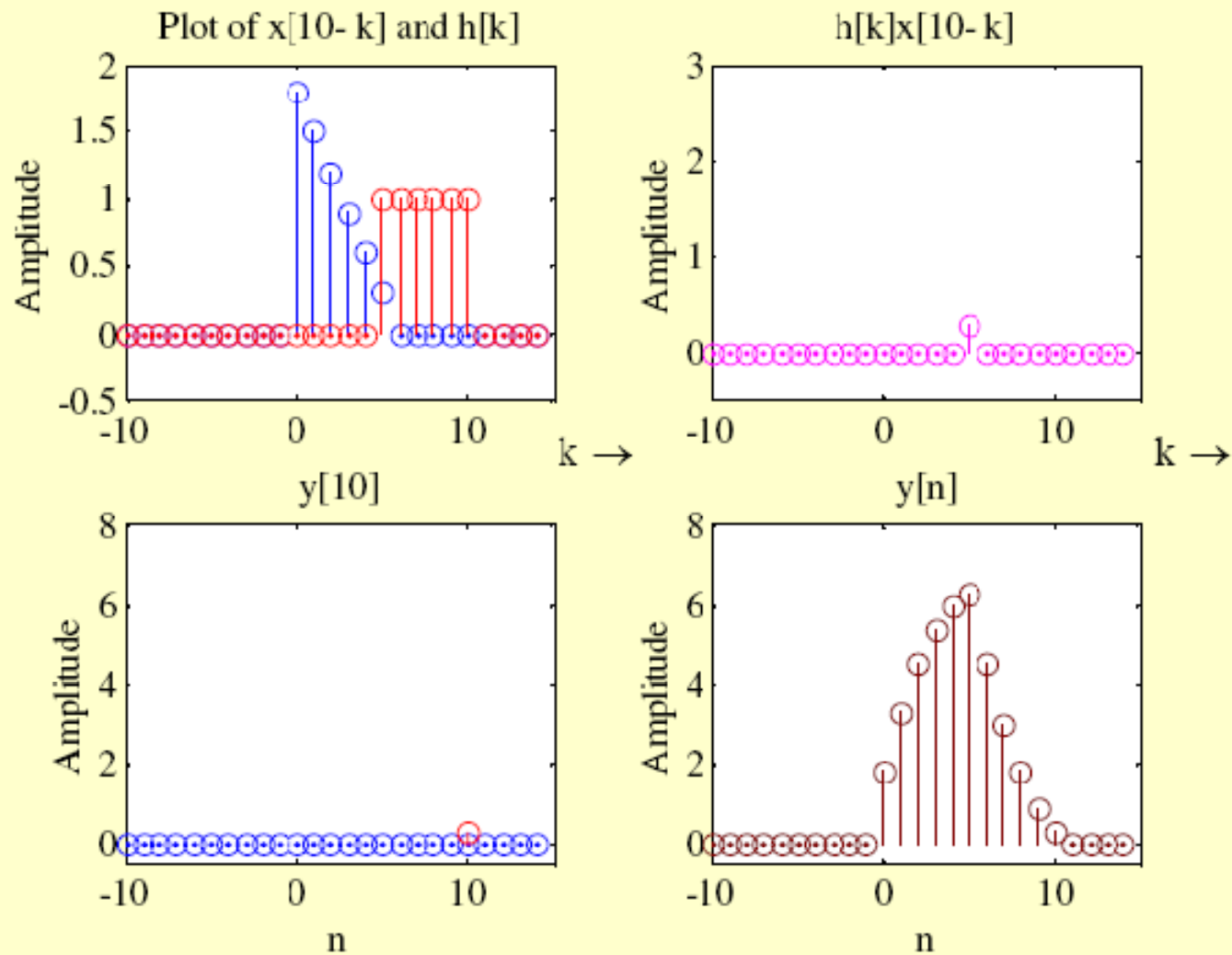
Convolution (8/12)



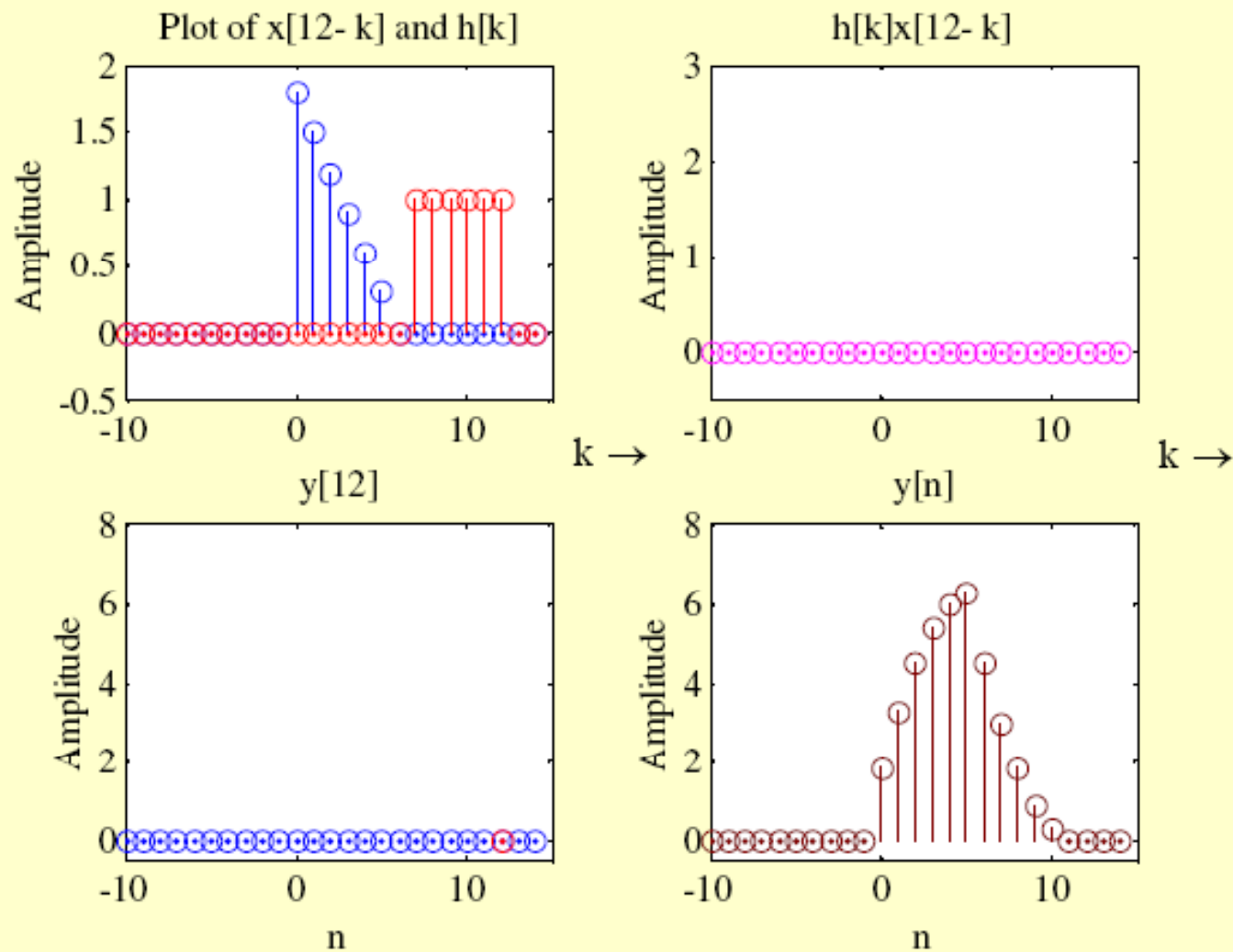
Convolution (9/12)



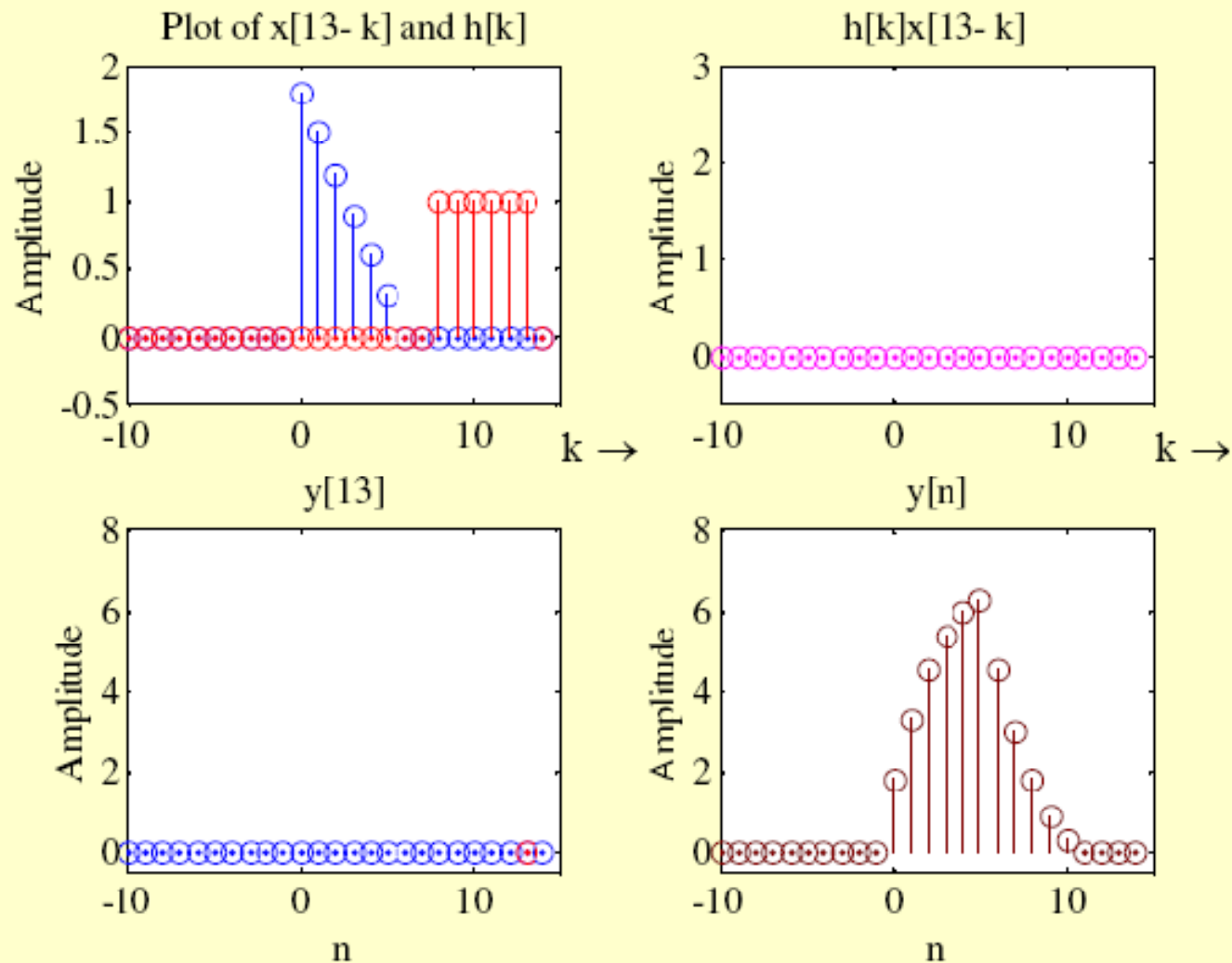
Convolution (10/12)



Convolution (11/12)



Convolution (12/12)

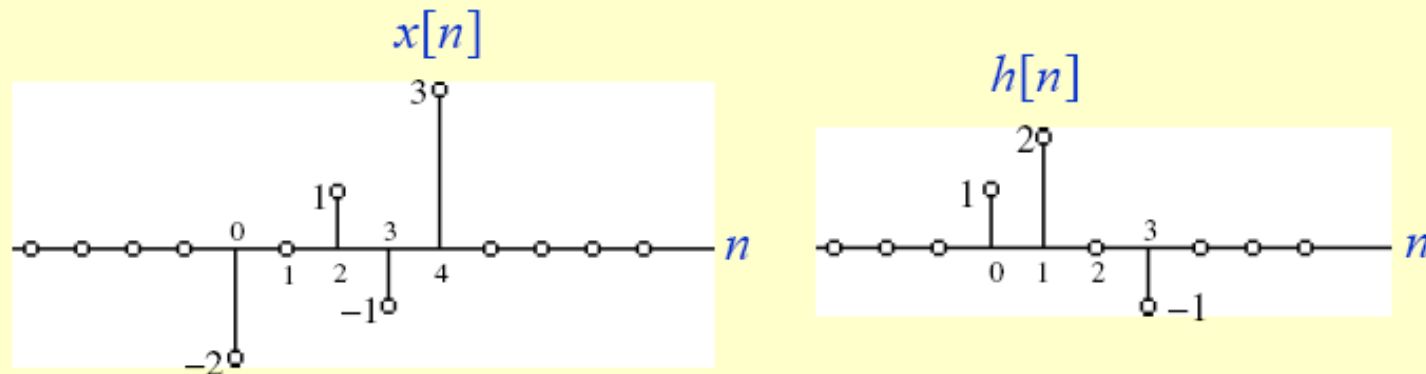


Computing Convolution Sum

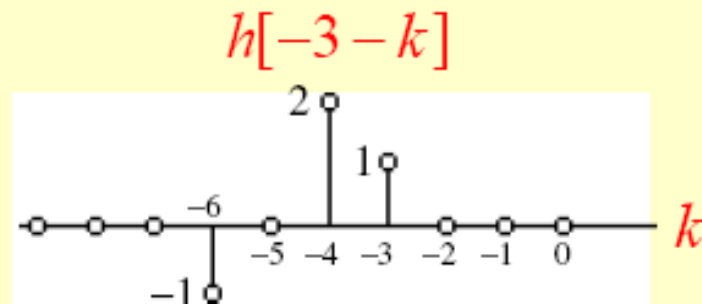
- In practice, if either the input or the impulse response is of finite length, the convolution sum can be used to compute the output sample as it involves a finite sum of products
- If both the input sequence and the impulse response sequence are of finite length, the output sequence is also of finite length
- If both the input sequence and the impulse response sequence are of infinite length, convolution sum cannot be used to compute the output

Convolution: Step by Step (1/4)

- Example - Develop the sequence $y[n]$ generated by the convolution of the sequences $x[n]$ and $h[n]$ shown below



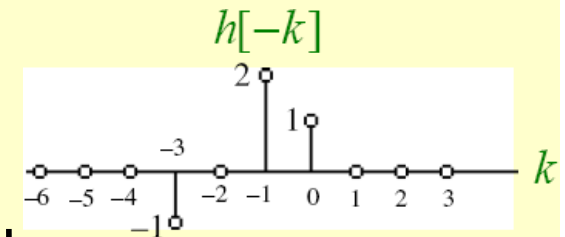
- As can be seen from the shifted time-reversed version $\{h[n - k]\}$ for $n < 0$, shown below for , for any value of the sample index k , the k -th sample of either $\{x[k]\}$ or $\{h[n - k]\}$ is zero



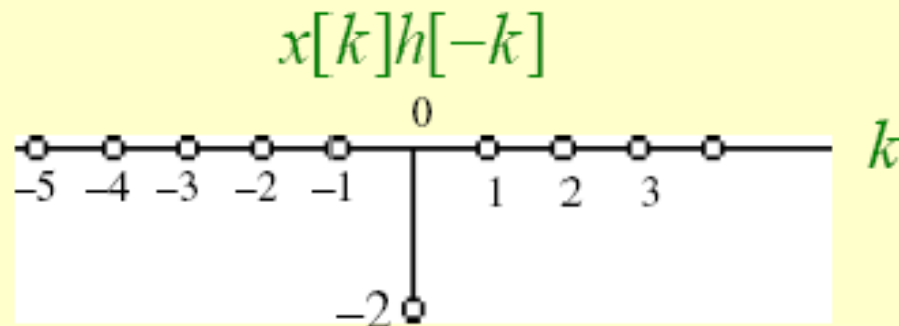
Convolution: Step by Step (2/4)

- As a result, for $n < 0$, the product of the k -th samples of $\{x[k]\}$ and $\{h[n - k]\}$ is always zero, and hence

$$y[n] = 0 \text{ for } n < 0$$

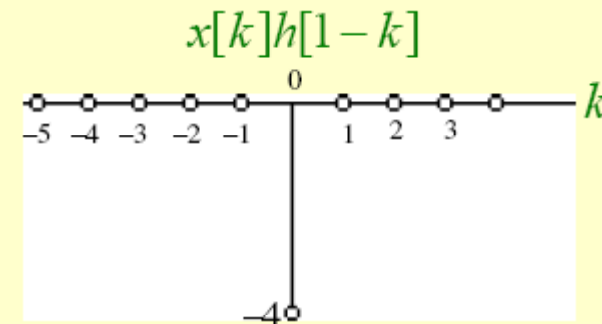
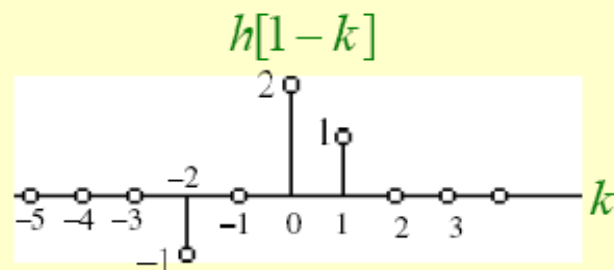


- Consider now the computation of $y[0]$
- The sequence $\{h[-k]\}$ is shown on the right
- The product sequence $\{x[k]h[-k]\}$ is plotted below which has a single nonzero sample $x[0]h[0]$ for $k = 0$
- Thus $y[0] = x[0]h[0] = -2$

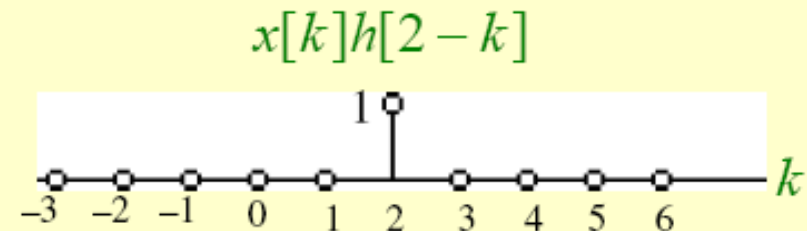
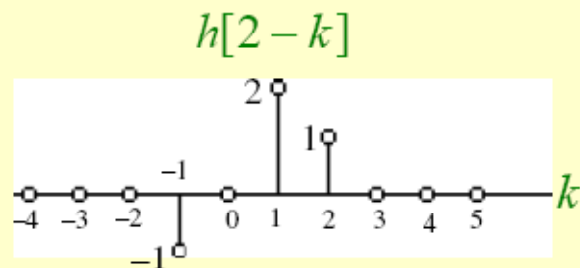


Convolution: Step by Step (3/4)

- For the computation of $y[1]$, we shift $\{h[-k]\}$ to the right by one sample period to form $\{h[1-k]\}$ as shown below
- Hence, $y[1] = x[0]h[1] + x[1]h[0] = -4 + 0 = -4$

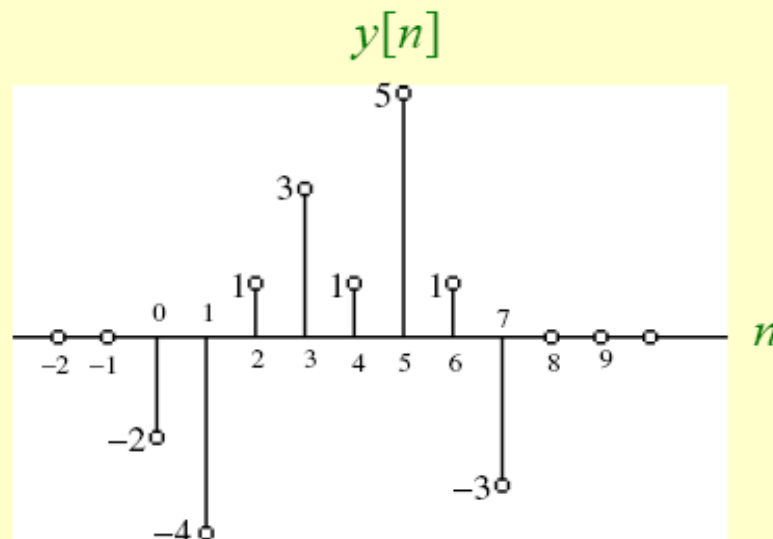


- Similarly, $y[2] = x[0]h[2] + x[1]h[1] + x[2]h[0] = 1$



Convolution: Step by Step (4/4)

- Repeat the process, we obtain the following output



- In general, if the lengths of the two sequences being convolved are M and N , then the sequence generated by the convolution is of length $M + N - 1$

Tabular Method of Convolution Sum Computation (1/2)

- Can be used to convolve two finite-length sequences

$n:$	0	1	2	3	4	5
$g[n]:$	$g[0]$	$g[1]$	$g[2]$	$g[3]$		
$h[n]:$	$h[0]$	$h[1]$	$h[2]$			
	$g[0]h[0]$	$g[1]h[0]$	$g[2]h[0]$	$g[3]h[0]$		
		$g[0]h[1]$	$g[1]h[1]$	$g[2]h[1]$	$g[3]h[1]$	
			$g[0]h[2]$	$g[1]h[2]$	$g[2]h[2]$	$g[3]h[2]$
$y[n]:$	$y[0]$	$y[1]$	$y[2]$	$y[3]$	$y[4]$	$y[5]$

$$y[n] = g[n] \circledast h[n]$$

$$y[0] = g[0]h[0]$$

$$y[1] = g[1]h[0] + g[0]h[1]$$

$$y[2] = g[2]h[0] + g[1]h[1] + g[0]h[2]$$

$$y[3] = g[3]h[0] + g[2]h[1] + g[1]h[2]$$

$$y[4] = g[3]h[1] + g[2]h[2]$$

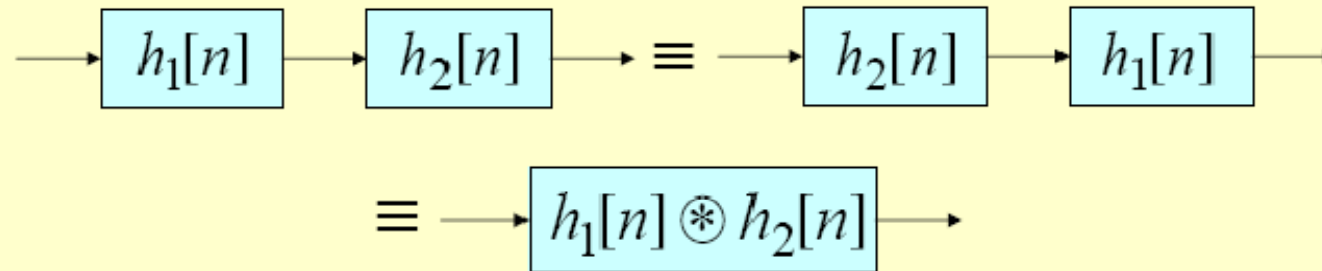
$$y[5] = g[3]h[2]$$

Tabular Method of Convolution Sum Computation (2/2)

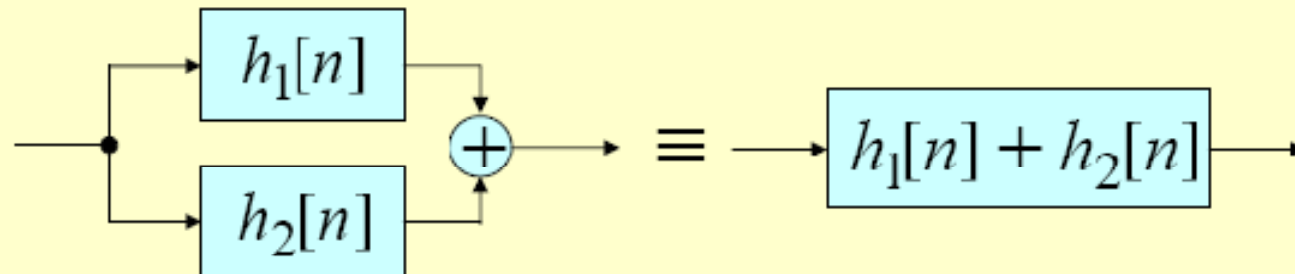
- The method can also be applied to convolve two finite-length two-sided sequences ◦
- In this case, a decimal point is first placed to the right of the sample with the time index $n = 0$ for each sequence
- Next, convolution is computed ignoring the location of the decimal point
- Finally, the decimal point is inserted according to the rules of conventional multiplication
- The sample immediately to the left of the decimal point is then located at the time index $n = 0$

Simple Interconnection Schemes

- Two simple interconnection schemes are:
 - Cascade Connection**



- Parallel Connection**

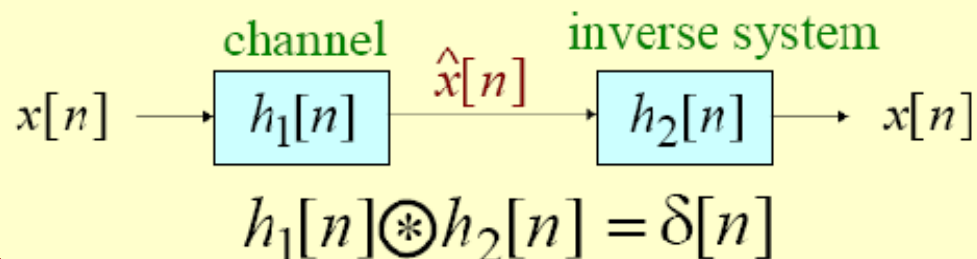


Cascade Connection (1/2)

- The ordering of the systems in the cascade has no effect on the overall impulse response
- A cascade connection of two stable systems is stable
- A cascade connection of two passive (lossless) systems is passive (lossless)
- An application is in the development of an **inverse system**
- If the cascade connection satisfies the relation

$$h_1[n] \circledast h_2[n] = \delta[n]$$

then the LTI system $h_1[n]$ is said to be the inverse of $h_2[n]$ and vice-versa



Cascade Connection (2/2)

- Example - Consider the discrete-time accumulator with an impulse response $\mu[n]$
- Its inverse system satisfy the condition

$$\mu[n] \circledast h_2[n] = \delta[n]$$

- It follows from the above that $h_2[n] = 0$ for $n < 0$ and

$$h_2[0] = 1$$
$$\sum_{\ell=0}^n h_2[\ell] = 0 \text{ for } n \geq 1$$

- Thus the impulse response of the inverse system of the discrete-time accumulator is given by

$$h_2[n] = \delta[n] - \delta[n-1] \text{ (backward difference system)}$$

Interconnection Schemes

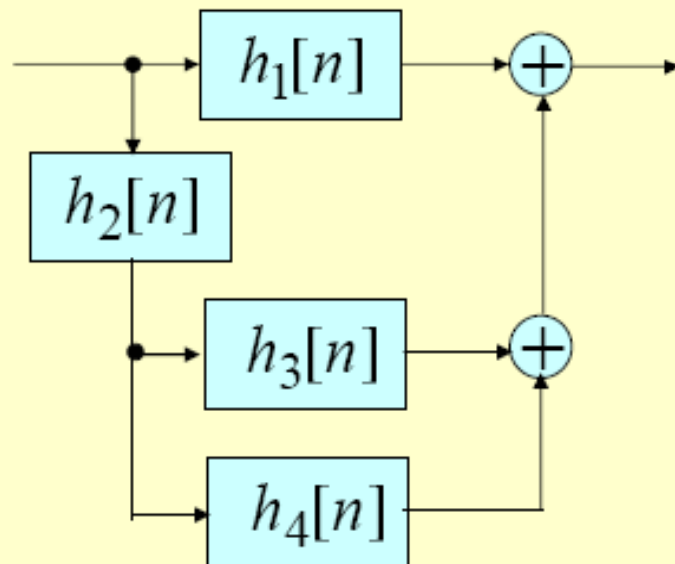
- Consider the discrete-time system where

$$h_1[n] = \delta[n] + 0.5\delta[n-1],$$

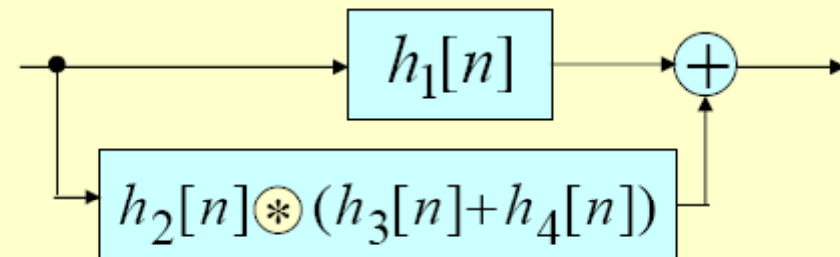
$$h_3[n] = 2\delta[n],$$

$$h_2[n] = 0.5\delta[n] - 0.25\delta[n-1],$$

$$h_4[n] = -2(0.5)^n \mu[n]$$



$\equiv$



$$\begin{aligned} h[n] &= h_1[n] + h_2[n] \otimes (h_3[n] + h_4[n]) \\ &= h_1[n] + h_2[n] \otimes h_3[n] + h_2[n] \otimes h_4[n] \end{aligned}$$

$$h[n] = \delta[n] + \frac{1}{2}\delta[n-1] + \delta[n] - \frac{1}{2}\delta[n-1] - \delta[n] = \delta[n]$$

Stability Condition of an LTI Discrete-Time System (1/3)

- **BIBO Stability Condition** - A discrete-time system is BIBO stable if and only if the output sequence $\{y[n]\}$ remains bounded for all bounded input sequence $\{x[n]\}$
- An LTI discrete-time system is BIBO stable if and only if its impulse response sequence $\{h[n]\}$ is absolutely summable, i.e.

$$S = \sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

- Proof: Assume $h[n]$ is a real sequence
- Since the input sequence $x[n]$ is bounded we have

therefore

$$|x[n]| \leq B_x < \infty$$
$$|y[n]| = \left| \sum_{k=-\infty}^{\infty} h[k]x[n-k] \right| \leq \sum_{k=-\infty}^{\infty} |h[k]| |x[n-k]| \leq B_x \sum_{k=-\infty}^{\infty} |h[k]| = B_x S$$

Stability Condition of an LTI Discrete-Time System (2/3)

- Thus, $S < \infty$ implies $|y[n]| \leq B_y < \infty$, indicating that $y[n]$ is also bounded
- To prove the converse, assume $y[n]$ is bounded, i.e., $|y[n]| \leq B_y$
- Consider the bounded input given by $x[n] = \text{sgn}(h[-n])$ where $\text{sgn}(c) = 1$, for $c \geq 0$, and $\text{sgn}(c) = -1$ for $c < 0$
- For this input, $y[n]$ at $n = 0$ is

$$y[0] = \sum_{k=-\infty}^{\infty} \text{sgn}(h[k])h[k] = S$$

- Therefore, if $S = \infty$, then $\{y[n]\}$ is not a bounded sequence

Stability Condition of an LTI Discrete-Time System (3/3)

- Example - Consider a causal LTI discrete-time system with an impulse response

$$h[n] = (\alpha)^n \mu[n]$$

- For this system

$$S = \sum_{n=-\infty}^{\infty} |\alpha^n \mu[n]| = \sum_{n=0}^{\infty} |\alpha|^n = \frac{1}{1-|\alpha|} \quad \text{if } |\alpha| < 1$$

- Therefore $S < \infty$ if $|\alpha| < 1$, for which the system is BIBO stable
- If $|\alpha| = 1$, the system is not BIBO stable

Causality Condition of an LTI Discrete-Time System (1/3)

- Let $x_1[n]$ and $x_2[n]$ be two input sequences with
 $x_1[n] = x_2[n]$ for $n \leq n_o$
- The corresponding output samples at of an LTI system with an impulse response $\{h[n]\}$ are then given by

$$\begin{aligned} y_1[n_o] &= \sum_{k=-\infty}^{\infty} h[k]x_1[n_o - k] = \sum_{k=0}^{\infty} h[k]x_1[n_o - k] \\ &\quad + \sum_{k=-\infty}^{-1} h[k]x_1[n_o - k] \\ y_2[n_o] &= \sum_{k=-\infty}^{\infty} h[k]x_2[n_o - k] = \sum_{k=0}^{\infty} h[k]x_2[n_o - k] \\ &\quad + \sum_{k=-\infty}^{-1} h[k]x_2[n_o - k] \end{aligned}$$

Causality Condition of an LTI Discrete-Time System (2/3)

- If the LTI system is also causal, then

$$y_1[n_o] = y_2[n_o]$$

- As $x_1[n] = x_2[n]$ for $n \leq n_o$

$$\sum_{k=0}^{\infty} h[k]x_1[n_o - k] = \sum_{k=0}^{\infty} h[k]x_2[n_o - k]$$

- This implies

$$\sum_{k=-\infty}^{-1} h[k]x_1[n_o - k] = \sum_{k=-\infty}^{-1} h[k]x_2[n_o - k]$$

- As for $x_1[n] \neq x_2[n]$ the only way the condition

$$\sum_{k=-\infty}^{-1} h[k]x_1[n_o - k] = \sum_{k=-\infty}^{-1} h[k]x_2[n_o - k] \quad \text{holds if } h[k] = 0 \text{ for } k < 0$$

Causality Condition of an LTI Discrete-Time System (3/3)

- An LTI discrete-time system is **causal** if and only if its impulse response $\{h[n]\}$ is a causal sequence ◦
- Example - The discrete-time accumulator defined by

$$y[n] = \sum_{\ell=-\infty}^n \delta[\ell] = \mu[n]$$

is causal as it has a causal impulse response

$$h[n] = \sum_{\ell=-\infty}^n \delta[\ell] = \mu[n]$$

- Example - The factor-of-2 interpolator defined by

$$y[n] = x_u[n] + \frac{1}{2}(x_u[n-1] + x_u[n+1])$$

is noncausal as it has a noncausal impulse response

$$\{h[n]\} = \{0.5 \quad 1 \quad 0.5\}$$

↑

Finite-Dimensional LTI Discrete-Time Systems

- An important subclass of LTI discrete-time systems is characterized by a linear constant coefficient difference equation of the form

$$\sum_{k=0}^N d_k y[n-k] = \sum_{k=0}^M p_k x[n-k]$$

where $\{d_k\}$ and $\{p_k\}$ are constants characterizing the system

- The **order** of the system is given by $\max(N, M)$, which is the order of the difference equation
- Suppose the system is causal, then the output $y[n]$ can be recursively computed using

$$y[n] = - \sum_{k=1}^N \frac{d_k}{d_0} y[n-k] + \sum_{k=0}^M \frac{p_k}{d_0} x[n-k] \quad \text{provided } d_0 \neq 0$$

Classification of LTI Discrete-Time Systems (1/3)

Based on Impulse Response Length -

- If the impulse response $h[n]$ is of finite length, i.e.,
 $h[n] = 0$ for $n < N_1$ and $n > N_2$, $N_1 < N_2$
then it is known as a **finite impulse response (FIR)** discrete-time system
- The convolution sum description here is

$$y[n] = \sum_{k=N_1}^{N_2} h[k]x[n-k]$$

- The output $y[n]$ of an FIR LTI discrete-time system can be computed directly from the convolution sum as it is a finite sum of products (e.g., moving-average filter & interpolator)

Classification of LTI Discrete-Time Systems (2/3)

- If the impulse response $h[n]$ is of infinite length, then it is known as a **infinite impulse response (IIR)** discrete-time system

- **Example** - The discrete-time accumulator defined by

$$y[n] = y[n-1] + x[n]$$

is an IIR system

- **Example** - The numerical integration formulas that are used to numerically solve integrals of the form

$$y(t) = \int_0^t x(\tau) d\tau$$

can be characterized by the following 1st-order IIR system

$$y[n] = y[n-1] + \frac{T}{2} \{x[n] + x[n-1]\}$$

Classification of LTI Discrete-Time Systems (3/3)

Based on the Output Calculation Process -

- **Nonrecursive System** - Here the output can be calculated sequentially, knowing only the present and past input samples
- **Recursive System** - Here the output computation involves past output samples in addition to the present and past input samples

Based on the Coefficients -

- **Real Discrete-Time System** - The impulse response samples are real valued
- **Complex Discrete-Time System** - The impulse response samples are complex valued

Correlation of Signals (1/4)

- There are applications where it is necessary to compare one reference signal with one or more signals
 - to determine the **similarity** between the pair
 - to determine **additional information based on the similarity**
- For example, in digital communications, the receiver has to determine which particular sequence has been received by comparing the received signal with possible sequences that may be transmitted
- Similarly, in radar and sonar applications, the received signal reflected from the target is a delayed (and even a distorted) version of the transmitted signal
- The received signal is often corrupted by additive random noise, making signal detection more complicated

Correlation of Signals (2/4)

- A measure of similarity between a pair of energy signals, $x[n]$ and $y[n]$, is given by the cross-correlation sequence

$$r_{xy}[\ell] = \sum_{n=-\infty}^{\infty} x[n]y[n-\ell], \quad \ell = 0, \pm 1, \pm 2, \dots$$

- The parameter ℓ called lag, indicates the time-shift between the pair of signals
- $y[n]$ is said to be shifted by ℓ samples to the right with respect to the reference sequence $x[n]$ for positive values of ℓ
- The ordering of the subscripts xy in $r_{xy}[\ell]$ specifies that $x[n]$ is the reference sequence which remains fixed in time while $y[n]$ is being shifted with respect to $x[n]$

Correlation of Signals (3/4)

- If $y[n]$ is made the reference signal and shift $x[n]$ with respect to $y[n]$, then the corresponding cross-correlation sequence is given by

$$\begin{aligned} r_{yx}[\ell] &= \sum_{n=-\infty}^{\infty} y[n]x[n-\ell] \\ &= \sum_{m=-\infty}^{\infty} y[m+\ell]x[m] = r_{xy}[-\ell] \end{aligned}$$

- The autocorrelation sequence of $x[n]$ is given by

$$r_{xx}[\ell] = \sum_{n=-\infty}^{\infty} x[n]x[n-\ell]$$

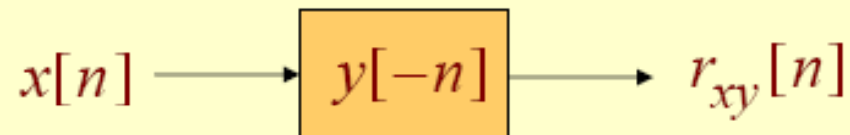
- Note, $r_{xx}[0] = \sum_{n=-\infty}^{\infty} x^2[n] = \mathcal{E}_x$, the energy of $x[n]$
- From the relation $r_{yx}[\ell] = r_{xy}[-\ell]$ it follows that $r_{xx}[\ell] = r_{xx}[-\ell]$, implying that $r_{xx}[\ell]$ is an even function for real $x[n]$

Correlation of Signals (4/4)

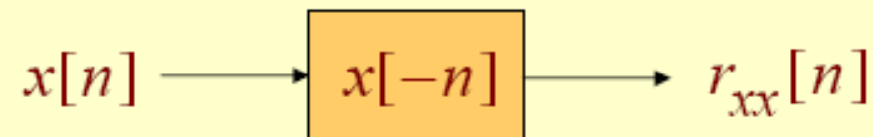
- Rewrite the expression for the cross-correlation as

$$r_{xy}[\ell] = \sum_{n=-\infty}^{\infty} x[n]y[-(\ell - n)] = x[\ell] \odot y[-\ell]$$

- The cross-correlation of $y[n]$ with the reference signal $x[n]$ can be computed by processing $x[n]$ with an LTI discrete-time system of impulse response $y[-n]$



- Likewise, the autocorrelation of $x[n]$ can be computed by processing $x[n]$ with an LTI discrete-time system of impulse response



Properties of Autocorrelation and Cross-correlation Sequences (1/3)

- Consider two finite-energy sequences $x[n]$ and $y[n]$
- The energy of the combined sequence $ax[n]+y[n-\ell]$ is also finite and nonnegative, i.e.,

$$\sum_{n=-\infty}^{\infty} (ax[n] + y[n-\ell])^2 = a^2 \sum_{n=-\infty}^{\infty} x^2[n] + 2a \sum_{n=-\infty}^{\infty} x[n]y[n-\ell] + \sum_{n=-\infty}^{\infty} y^2[n-\ell] \geq 0$$

- Thus

$$a^2 r_{xx}[0] + 2a r_{xy}[\ell] + r_{yy}[0] \geq 0$$

where $r_{xx}[0] = E_x > 0$ and $r_{yy}[0] = E_y > 0$

- The above equation can be rewritten as

$$\begin{bmatrix} a & 1 \end{bmatrix} \begin{bmatrix} r_{xx}[0] & r_{xy}[\ell] \\ r_{xy}[\ell] & r_{yy}[0] \end{bmatrix} \begin{bmatrix} a \\ 1 \end{bmatrix} \geq 0 \quad \text{for any finite value of } a$$

Properties of Autocorrelation and Cross-correlation Sequences (2/3)

- The matrix $\begin{bmatrix} r_{xx}[0] & r_{xy}[\ell] \\ r_{xy}[\ell] & r_{yy}[0] \end{bmatrix}$

is thus positive semidefinite

$$\Rightarrow r_{xx}[0]r_{yy}[0] - r_{xy}^2[\ell] \geq 0$$

or, equivalently,

$$|r_{xy}[\ell]| \leq \sqrt{r_{xx}[0]r_{yy}[0]} = \sqrt{\mathcal{E}_x \mathcal{E}_y}$$

- The inequality provides an upper bound for the cross-correlation samples
- If we set $y[n] = x[n]$, then the inequality reduces to

$$|r_{xx}[\ell]| \leq r_{xx}[0] = E_x$$

Properties of Autocorrelation and Cross-correlation Sequences (3/3)

- Thus, at zero lag ($l = 0$), the sample value of the autocorrelation sequence has its maximum value
- Now consider the case

$$y[n] = \pm b x[n - N]$$

where N is an integer and $b > 0$ is an arbitrary number

- In this case $\mathcal{E}_y = b^2 \mathcal{E}_x$
- Therefore

$$\sqrt{\mathcal{E}_x \mathcal{E}_y} = \sqrt{b^2 \mathcal{E}_x^2} = b \mathcal{E}_x$$

- Using the above result:

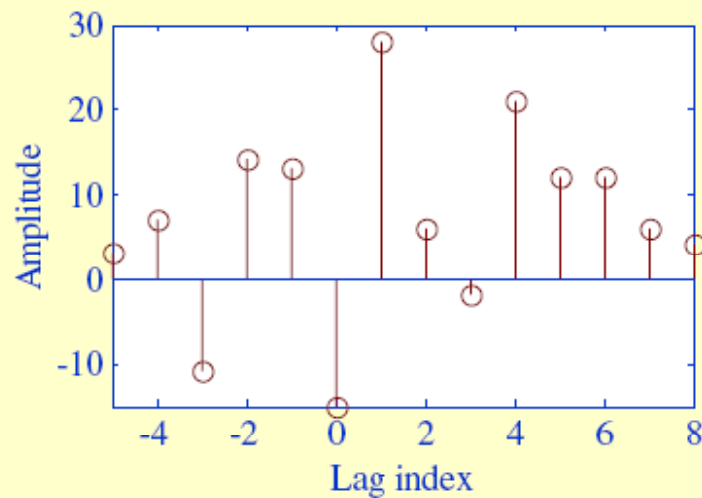
$$|r_{xy}[\ell]| \leq \sqrt{r_{xx}[0] r_{yy}[0]} = \sqrt{\mathcal{E}_x \mathcal{E}_y}$$

- We get $-br_{xx}[0] \leq r_{xy}[l] \leq br_{xx}[0]$

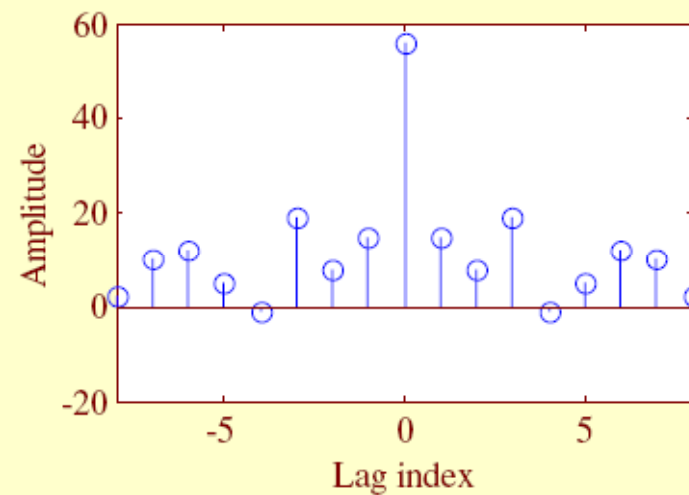
Computation of Correlations (1/2)

- Example - Consider the two finite-length sequences

$$x[n] = [1 \ 3 \ -2 \ 1 \ 2 \ -1 \ 4 \ 4 \ 2], \ y[n] = [2 \ -1 \ 4 \ 1 \ -2 \ 3]$$



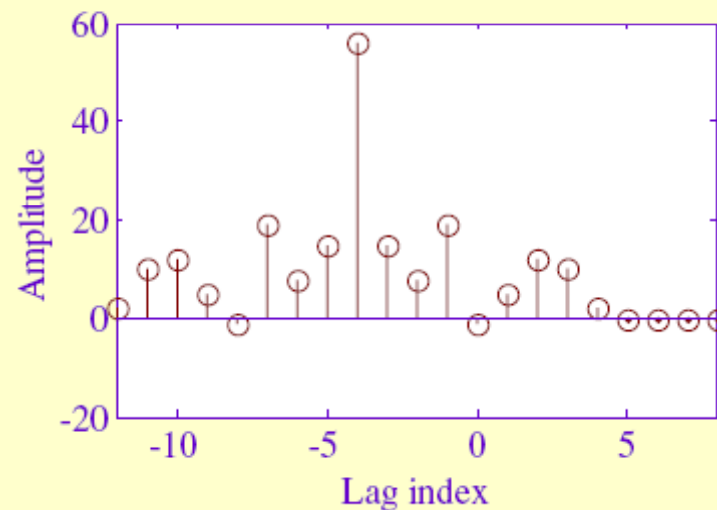
$$r_{xy}[n]$$



$$r_{xx}[n]$$

Computation of Correlations (2/2)

- Example - The cross-correlation of $x[n]$ and $y[n] = x[n - N]$ for $N = 4$
- Note: The peak of the cross-correlation is precisely the value of the delay N



$$r_{xy}[n]$$

Normalized Forms of Correlation

- Normalized forms of autocorrelation and cross-correlation are given by

$$\rho_{xx}[\ell] = \frac{r_{xx}[\ell]}{r_{xx}[0]}, \quad \rho_{xy}[\ell] = \frac{r_{xy}[\ell]}{\sqrt{r_{xx}[0]r_{yy}[0]}}$$

- Note: $|\rho_{xx}[\ell]| \leq 1$ and $|\rho_{yy}[\ell]| \leq 1$ independent of the range of values of $x[n]$ and $y[n]$
- The cross-correlation sequence for a pair of power signals, $x[n]$ and $y[n]$, is defined as

$$r_{xy}[\ell] = \lim_{K \rightarrow \infty} \frac{1}{2K+1} \sum_{n=-K}^K x[n]y[n-\ell]$$

- The autocorrelation sequence of a power signal $x[n]$ is given by

$$r_{xx}[\ell] = \lim_{K \rightarrow \infty} \frac{1}{2K+1} \sum_{n=-K}^K x[n]x[n-\ell]$$

Normalized Forms of Correlation

- The cross-correlation sequence for a pair of periodic signals of period N , $\tilde{x}[n]$ and $\tilde{y}[n]$, is given by

$$r_{\tilde{x}\tilde{y}}[\ell] = \frac{1}{N} \sum_{n=0}^{N-1} \tilde{x}[n] \tilde{y}[n - \ell]$$

- The autocorrelation sequence of $\tilde{x}[n]$:

$$r_{\tilde{x}\tilde{x}}[\ell] = \frac{1}{N} \sum_{n=0}^{N-1} \tilde{x}[n] \tilde{x}[n - \ell]$$

- Both $r_{\tilde{x}\tilde{y}}[\ell]$ and $r_{\tilde{x}\tilde{x}}[\ell]$ are also periodic with a period N
- Let $\tilde{x}[n]$ be a periodic signal corrupted by the random noise $d[n]$ resulting in the signal

$$w[n] = \tilde{x}[n] + d[n]$$

which is observed for $0 \leq n \leq M - 1$ where $M \gg N$

Normalized Forms of Correlation

- The autocorrelation of $w[n]$ is given by

$$\begin{aligned}
 r_{ww}[\ell] &= \frac{1}{M} \sum_{n=0}^{M-1} w[n]w[n-\ell] \\
 &= \frac{1}{M} \sum_{n=0}^{M-1} (\tilde{x}[n] + d[n])(\tilde{x}[n-\ell] + d[n-\ell]) \\
 &= \frac{1}{M} \sum_{n=0}^{M-1} \tilde{x}[n]\tilde{x}[n-\ell] + \frac{1}{M} \sum_{n=0}^{M-1} d[n]d[n-\ell] \\
 &\quad + \frac{1}{M} \sum_{n=0}^{M-1} \tilde{x}[n]d[n-\ell] + \frac{1}{M} \sum_{n=0}^{M-1} d[n]\tilde{x}[n-\ell] \\
 &= r_{\tilde{x}\tilde{x}}[\ell] + r_{dd}[\ell] + r_{\tilde{x}d}[\ell] + r_{d\tilde{x}}[\ell]
 \end{aligned}$$

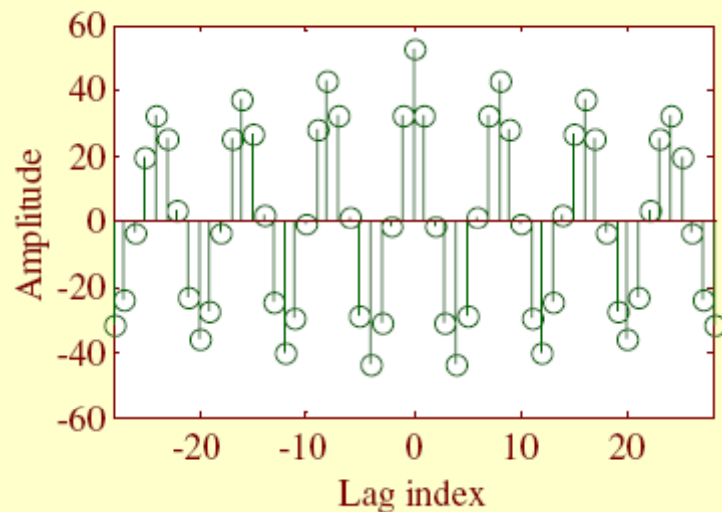
- $r_{\tilde{x}\tilde{x}}[\ell]$ is a periodic sequence with a period N and hence will have peaks at $\ell = 0, N, 2N, \dots$ with the same amplitudes as ℓ approaches M
- As $\tilde{x}[n]$ and $d[n]$ are not correlated, samples of cross-correlation sequences and $r_{\tilde{x}d}[\ell]$ are $r_{d\tilde{x}}[\ell]$ likely to be very small relative to the amplitudes of $r_{\tilde{x}\tilde{x}}[\ell]$

Normalized Forms of Correlation

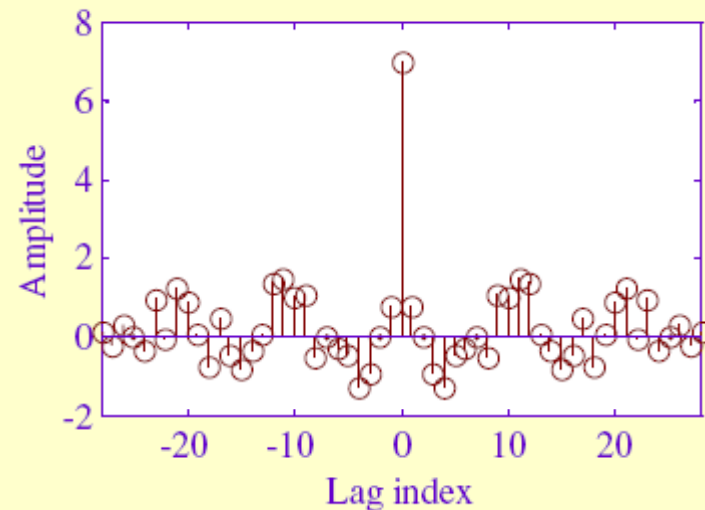
- The autocorrelation $r_{dd}[l]$ of $d[n]$ will show a peak at $l = 0$ with other samples having rapidly decreasing amplitudes with increasing values of $|l|$
- Hence, peaks of $r_{ww}[l]$ for $l > 0$ are essentially due to the peaks of $r_{\tilde{x}\tilde{x}}[l]$ and can be used to determine whether $\tilde{x}[n]$ is a periodic sequence and also its period N if the peaks occur at periodic intervals

Normalized Forms of Correlation

- Example - Determine the period of the sinusoidal sequence $x[n] = \cos(0.25n)$, $0 \leq n \leq 95$ corrupted by an additive uniformly distributed random noise of amplitude in the range $[-0.5, 0.5]$



$$r_{ww}[l]$$



$$r_{dd}[l]$$